



# PREHOSPITAL EMERGENCY CARE CLINICAL PROTOCOLS



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## Preface

These Clinical Protocols are to be used by trained and competent professionals as a guide of evidence-based best practice for out-of-hospital emergency care in developing health systems. They are not a substitute for training and should only be used by those specifically trained on the procedures documented within. This document is not intended to be a training tool or textbook. This document is reviewed and updated every two years, whilst open access with no copyright restrictions, if used, it is the user's responsibility to ensure that these protocols are the most up-to-date. Every effort has been made to ensure this document is free from errors. If one is found, please contact the author, who will ensure an update is released.

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## General Definitions of Titles Used and Scope of Practice

This document refers to the titles “**Emergency Medical Technician (EMT)**” and “**Paramedic**”. The titles do not correlate with titles used in other areas such as the UK, USA or Australia, but rather, they align with the WHO curriculum, current region-specific scope of practice, expert consensus statement and evidence-based practice. This is due to the unique demands of healthcare providers in developing healthcare systems where resources, topography and clinical presentations vary greatly from those in developed systems. Below is a general outline of the recommended scope of each, with reference to expected skills and interventions.

| Intervention                                                              | Emergency Medical Technician (EMT) | Paramedic |
|---------------------------------------------------------------------------|------------------------------------|-----------|
| Act as the lead healthcare provider in an Ambulance                       |                                    |           |
| ABCDE Assessment                                                          |                                    |           |
| Basic Airway Management (OPA, NPA, Airway Manoeuvres)                     |                                    |           |
| Supraglottic Airway (I Gel, LMA, King Tube)                               |                                    |           |
| Endotracheal Intubation                                                   | Not within scope of practice       |           |
| BVM Ventilation                                                           |                                    |           |
| Oxygen Delivery                                                           |                                    |           |
| Suction                                                                   |                                    |           |
| IV/ IO Cannulation                                                        | Not within scope of practice       |           |
| Laryngoscopy                                                              | Not within scope of practice       |           |
| 12 Lead ECG Acquisition and Interpretation                                |                                    |           |
| Transcutaneous Cardiac Pacing                                             | Not within scope of practice       |           |
| Undertake all Vital Signs and interpret findings.                         |                                    |           |
| Undertake Trauma Assessment                                               |                                    |           |
| Manage Major Trauma Patient                                               |                                    |           |
| Manage Normal Childbirth                                                  |                                    |           |
| Manage Obstetrical Emergencies and the critically ill Obstetrics patient. | Not within scope of practice       |           |
| Administer IM / Oral Medication                                           | Subject to local Medical Direction |           |
| Administer IV Medications (Excluding NACL/D10)                            | Not within scope of practice       |           |
| Administer IO Medications                                                 | Not within scope of practice       |           |
| Administer NACL/D10 IV                                                    | Subject to local Medical Direction |           |
| Administer Tranexamic Acid (IM/IV)                                        | Subject to local Medical Direction |           |
| Confirmation of Death                                                     | Subject to local Medical Direction |           |
| Withholding Resuscitation in an obvious death                             | Subject to local Medical Direction |           |

## Clinical Protocols Structure

All protocols in this document follow the same, uniform structure, as detailed below. They have been created to conform to AGREE II criteria for clinical guidelines.

### Purpose

#### 1. Scope and Definitions

- 1.1 Scope
- 1.2 Definitions

#### 2. Assessment

- 2.1 Initial Assessment
- 2.2 Focused Assessment
- 2.3 Vital Signs and Monitoring

#### 3. Management

- 3.1 **EMT-Level Management +**
- 3.2 Paramedic-Level Management
- 3.3 Special Considerations (if applicable)

#### 4. Key Points



## Glossary of Abbreviations

|                  |   |                                                      |
|------------------|---|------------------------------------------------------|
| <b>ABCDE</b>     | – | Airway, Breathing, Circulation, Disability, Exposure |
| <b>ACS</b>       | – | Acute Coronary Syndrome                              |
| <b>AED</b>       | – | Automated External Defibrillator                     |
| <b>ALS</b>       | – | Advanced Life Support                                |
| <b>APH</b>       | – | Antepartum Haemorrhage                               |
| <b>AVPU</b>      | – | Alert, Voice, Pain, Unresponsive                     |
| <b>AGP</b>       | – | Aerosol Generating Procedure                         |
| <b>BGL</b>       | – | Blood Glucose Level                                  |
| <b>BP</b>        | – | Blood Pressure                                       |
| <b>BVM</b>       | – | Bag-Valve Mask                                       |
| <b>CABG</b>      | – | Coronary Artery Bypass Graft                         |
| <b>CAD</b>       | – | Coronary Artery Disease                              |
| <b>CNS</b>       | – | Central Nervous System                               |
| <b>CO</b>        | – | Carbon Monoxide                                      |
| <b>CPAP</b>      | – | Continuous Positive Airway Pressure                  |
| <b>CSF</b>       | – | Cerebrospinal Fluid                                  |
| <b>CT</b>        | – | Computed Tomography                                  |
| <b>CVA</b>       | – | Cerebrovascular Accident (Stroke)                    |
| <b>C-Section</b> | – | Caesarean Section                                    |
| <b>DOB</b>       | – | Date of Birth                                        |
| <b>DTP</b>       | – | Drug therapy protocol                                |
| <b>DVT</b>       | – | Deep Vein Thrombosis                                 |
| <b>ECG</b>       | – | Electrocardiogram                                    |

|                         |   |                                                                              |
|-------------------------|---|------------------------------------------------------------------------------|
| <b>ED</b>               | – | Emergency Department                                                         |
| <b>EFAST</b>            | – | Extended Focused Assessment with Sonography for Trauma                       |
| <b>EMT</b>              | – | Emergency Medical Technician                                                 |
| <b>ETCO<sub>2</sub></b> | – | End-Tidal Carbon Dioxide                                                     |
| <b>BEFAST</b>           | – | Balance, Eyes, Face, Arms, Speech, Time (Stroke screening)                   |
| <b>FiO<sub>2</sub></b>  | – | Fraction of Inspired Oxygen                                                  |
| <b>GCS</b>              | – | Glasgow Coma Scale                                                           |
| <b>GTN</b>              | – | Glyceryl Trinitrate                                                          |
| <b>HOTT</b>             | – | Hypovolaemia, Oxygenation failure, Tension Pneumothorax, Tamponade (cardiac) |
| <b>HR</b>               | – | Heart Rate                                                                   |
| <b>ICP</b>              | – | Intracranial Pressure                                                        |
| <b>IM</b>               | – | Intramuscular                                                                |
| <b>IO</b>               | – | Intraosseous                                                                 |
| <b>IV</b>               | – | Intravenous                                                                  |
| <b>LOC</b>              | – | Loss of Consciousness                                                        |
| <b>LMA</b>              | – | Laryngeal Mask Airway                                                        |
| <b>L/min</b>            | – | Litres per Minute                                                            |
| <b>MAP</b>              | – | Mean Arterial Pressure                                                       |
| <b>MI</b>               | – | Myocardial Infarction                                                        |
| <b>NIBP</b>             | – | Non-Invasive Blood Pressure                                                  |
| <b>NPA</b>              | – | Nasopharyngeal Airway                                                        |
| <b>NSAID</b>            | – | Non-Steroidal Anti-Inflammatory Drug                                         |
| <b>NSTEMI</b>           | – | Non-ST-Elevation Myocardial Infarction                                       |
| <b>NIV</b>              | – | Non-Invasive Ventilation                                                     |
| <b>O<sub>2</sub></b>    | – | Oxygen                                                                       |

|                        |   |                                                                                                                                    |
|------------------------|---|------------------------------------------------------------------------------------------------------------------------------------|
| <b>OPA</b>             | – | Oropharyngeal airway                                                                                                               |
| <b>OPQRST</b>          | – | Onset, Provocation, Quality, Radiation, Severity, Time                                                                             |
| <b>PALS</b>            | – | Paediatric Advanced Life Support                                                                                                   |
| <b>PE</b>              | – | Pulmonary Embolism                                                                                                                 |
| <b>PPE</b>             | – | Personal Protective Equipment                                                                                                      |
| <b>PEEP</b>            | – | Positive End-Expiratory Pressure                                                                                                   |
| <b>PMH</b>             | – | Past Medical History                                                                                                               |
| <b>PPH</b>             | – | Post-Partum Haemorrhage                                                                                                            |
| <b>PPE</b>             | – | Personal Protective Equipment                                                                                                      |
| <b>PPCI</b>            | – | Primary Percutaneous Coronary Intervention                                                                                         |
| <b>RR</b>              | – | Respiratory Rate                                                                                                                   |
| <b>SAMPLE</b>          | – | Signs and Symptoms, Allergies, Medications, Past Medical History, Last Oral Intake, Events Leading Up to Present Illness or Injury |
| <b>SOB</b>             | – | Shortness of Breath                                                                                                                |
| <b>STEMI</b>           | – | ST-Elevation Myocardial Infarction                                                                                                 |
| <b>SVT</b>             | – | Supraventricular Tachycardia                                                                                                       |
| <b>TIA</b>             | – | Transient Ischaemic Attack                                                                                                         |
| <b>Tx</b>              | – | Treatment                                                                                                                          |
| <b>Tranexamic Acid</b> | – | Tranexamic Acid                                                                                                                    |
| <b>UTI</b>             | – | Urinary Tract Infection                                                                                                            |
| <b>VF</b>              | – | Ventricular Fibrillation                                                                                                           |
| <b>VT</b>              | – | Ventricular Tachycardia                                                                                                            |

1



Medical and Surgical

**M1 CARDIAC ARREST**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for assessing and managing patients with cardiac arrest in the prehospital setting.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing patients in cardiac arrest from all causes.

#### 1.2 Definitions

- **Cardiac Arrest:** The sudden cessation of effective cardiac activity, resulting in the absence of signs of circulation such as responsiveness, normal breathing, and a palpable pulse.
- **Witnessed Cardiac Arrest:** Cardiac arrest observed by ambulance staff or healthcare providers only – not the general public or family members.
- **Resuscitation:** The application of CPR, defibrillation, and advanced life support measures to restore circulation.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure the safety of providers and the patient.
- **Primary Survey (ABCDE):**
  - **Airway:** Basic manoeuvres and/or OPA/IGEL/LMA, suction or position for drainage if required.
  - **Breathing:** Initiate ventilations as per BLS/ALS guidelines with high-flow oxygen with a bag valve mask. If drowning or paediatric, deliver 5 rescue breaths before CPR.
  - **Circulation:** Immediate high-quality CPR
  - **Defibrillation:** Immediately attach AED/Manual defibrillator to patient and assess rhythm – In adult patients, application of AED is prioritised over CPR if a single healthcare provider.

### 2.2 Focused Assessment

- Use 4 H's and 4 T's to guide focused assessment, whilst resuscitation is ongoing
- Consider and address where possible: Hypoxia, Hypovolemia, Hyper/Hypokalaemia, Hypothermia, Toxins, Tamponade, Tension Pneumothorax and Thrombosis.

### 2.3 Vital Signs and Monitoring

- The only essential continuous vital signs monitoring, if available, is end-tidal CO2 monitoring during cardiac arrest. If ROSC is achieved, a full assessment and vital signs should be conducted immediately. When considering H's and T's, Temperature and Blood glucose should be checked early in the cardiac arrest so that if they are deranged, efforts can be made to correct them.

### 3. Management

#### 3.1 EMT-Level Management

- Immediate CPR: 30 chest compressions to 2 ventilations.
- Attach the Automated External Defibrillator (AED) and follow instructions.
- Bag-Valve-Mask (BVM) ventilation with oropharyngeal airway.
- Request Advanced Life Support (ALS) trained staff.
- Continue resuscitation for 20 minutes, decide based on location and resources if transport to hospital is reasonable, or consideration would be given to terminate resuscitative efforts after 20 minutes by utilising local medical direction if no response to treatment.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Immediate BLS interventions.
- Assess rhythm (shockable/non-shockable).
- Use capnography where available.
- Continuous compressions if an advanced airway is present.
- Establish intravenous (IV) or intraosseous (IO) access; there is no preference for a first-line intervention.
- Administer Adrenaline 1mg every 3-5 minutes.
- Administer Amiodarone after the 3rd shock (300mg) for shockable rhythms and after the 5th shock (150mg).
- Consider reversible causes (4 Hs and 4 Ts).
- If Return of Spontaneous Circulation (ROSC) occurs: Use the ABCDE approach and obtain a 12-lead ECG. Consider push dose adrenaline for circulatory support.
- Transport with resuscitation ongoing is not recommended for the Paramedic scope unless the patient is pregnant, a renal dialysis patient, paediatric or poisoning/overdose.



#### 4. Key Points

- Withholding resuscitation would be appropriate in the following circumstances:
  - Injuries incompatible with life – Decapitation, hemicorporectomy, 100% burns, brain matter showing.
  - There are irreversible signs of death: Rigour mortis, decomposition, dependent lividity.
  - No resuscitation has been commenced within 15 minutes of recognition of cardiac arrest.
  - There is a danger to the responders to initiate resuscitation.
  - If the patient has an advanced, end-stage disease such as COPD, liver failure, cancer or dementia, consult medical direction urgently to discuss termination or non-commencement of resuscitation. Do not delay initiation of resuscitation to contact medical direction.
- If ROSC is achieved, the patient should be transported to the nearest emergency department for stabilisation.
- Traumatic Cardiac Arrest: This demands unique and aggressive management with differing priorities.
- Use **HOTT Protocol** (Traumatic Cardiac Arrest Reversible Causes) and consider/address ALL steps PRIOR to commencing CPR/defibrillation.
  - **H** – **Hypovolemia** (due to major blood loss) - Consider arterial tourniquets, pelvic binder, direct pressure and Tranexamic Acid.
  - **O** – **Oxygenation failure** (e.g., hypoxia, airway obstruction) - Provide high-flow oxygen and advanced airway management where possible.
  - **T** – **Tension pneumothorax** - All traumatic cardiac arrest patients should have bilateral needle chest decompression immediately, where available, chest drain placement is indicated.
  - **T** – **Tamponade (cardiac)** - Unable to correct prehospital, if suspected, consider immediate rapid transport.



**M2****AIRWAY OBSTRUCTION**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for assessing and managing patients with airway obstruction in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing patients of all ages with airway obstruction.

#### 1.2 Definitions

- **Airway Obstruction:** The obstruction of any part of the airway due to a foreign body or pathology that impacts the ability of a patient to breathe normally.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure the safety of providers and the patient.
- **Primary Survey (ABCDE):**
  - **Airway:** Assess for partial or full airway obstruction.
  - **Partial:** Patient may be able to talk, cough or may have stridor present
  - **Full:** Patient will be unable to talk, no audible sounds present and will result in cardiorespiratory arrest rapidly.
  - Correcting the airway obstruction is of the highest importance, and therefore, you will not be able to proceed with the rest of the ABCDE assessment until this is done, as it will rapidly lead to respiratory and cardiac arrest. Once airway obstruction is removed, proceed with;
  - **Breathing:** Assess adequacy of ventilation and oxygenation.
  - **Circulation:** Check for signs of shock (e.g., pale, cool, clammy skin; weak pulse).

### 2.2 Focused Assessment

- Not applicable until after the obstruction is removed, then follow standard history taking.

### 2.3 Vital Signs and Monitoring

- Prepare for cardiorespiratory arrest if unable to remove the obstruction
- Record SPO2, respiratory rate, blood pressure, heart rate, temperature and blood glucose after removing the obstruction.

### 3. Management

#### 3.1 EMT-Level Management

- **Conscious Adult/Child:**
  - Encourage effective coughing.
  - If the cough is ineffective:
    - Perform 5 back blows followed by 5 abdominal thrusts (Heimlich manoeuvre) for adults and children.
    - For pregnant or obese patients, use chest thrusts instead of abdominal thrusts.
    - Repeat until dislodged.
- **Infant (<1 year old):**
  - Deliver 5 back blows followed by 5 chest thrusts. Repeat until dislodged.
- **Unresponsive Patient:**
  - Begin CPR immediately, starting with chest compressions.
  - Check the airway during compressions and remove any visible obstruction (avoid blind sweeps).
  - If unable to visualise and remove the obstruction, consider attempting to ventilate as part of CPR to provide oxygen.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Prepare for advanced airway management with endotracheal intubation.
- Consider needle cricothyroidotomy if unable to dislodge the obstruction and ventilate.

#### 4. Key Points

- For pregnant patients, use chest thrusts rather than abdominal thrusts.
- If unable to dislodge, consider rapid transport to the closest emergency department.
- Needle cricothyroidotomy can be a bridge to a surgical airway in complete obstruction.

**M3****ANAPHYLAXIS**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for recognising and managing anaphylaxis in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing anaphylaxis in the prehospital setting.

#### 1.2 Definition

A severe, life-threatening, systemic hypersensitivity reaction characterised by rapid onset and involving multiple organ systems. It typically includes airway compromise, breathing difficulty, and circulatory collapse. Common triggers include foods, insect stings, and medications.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure safety for providers and patients.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions. If stridor or wheeze is present, administer 0.5mg IM adrenaline immediately in the anterolateral thigh.
  - **Breathing:** Assess adequacy of ventilation.
  - **Circulation:** Check pulse and perfusion.

### 2.2 Focused Assessment

- **History (SAMPLE): +**
  - Events leading up to anaphylaxis: Identify possible triggers.
- **Physical Examination:**
  - Inspect for airway swelling or compromise.
  - Assess for cutaneous symptoms such as urticaria, angioedema or itching.
  - Monitor for neurological signs (e.g., altered mental status).

### 2.3 Vital Signs and Monitoring

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

- Immediate IM adrenaline 0.5mg in the anterolateral thigh, repeat every 5 minutes if respiratory distress continues.
- Salbutamol, 5mg if severe wheeze present
- Rapid transport to the emergency department with the patient in the supine position

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- IV fluid resuscitation, 500ml bolus
- IV corticosteroids such as hydrocortisone 100mg
- Prepare for circulatory collapse.
- Rapid transport

### 4. Key Points

- Any systemic symptoms such as tachycardia or cutaneous symptoms + respiratory distress is anaphylaxis. A patient may have severe cutaneous symptoms but no respiratory distress, this is not anaphylaxis, and inversely, a patient may simply have tachycardia and respiratory distress with a history of an allergen exposure- this is anaphylaxis.
- Anaphylaxis patients regularly require large volumes of IV fluids.

**M4****RESPIRATORY DISTRESS**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for recognising and managing respiratory distress in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing respiratory distress in the prehospital setting.

#### 1.2 Definition

A blanket term is used for patients with subjective or objective difficulty breathing due to an undifferentiated pathology. The most common causes of respiratory distress that must be considered are;

- Asthma
- COPD
- Anaphylaxis
- Acute Cardiogenic Pulmonary Oedema
- Anxiety
- Tuberculosis
- Sepsis



## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure safety for providers and patients.
- **Primary Survey (ABCDE):** - Identify and address immediate life-threatening conditions.
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation.
  - **Circulation:** Check pulse and perfusion.

### 2.2 Focused Assessment

- **History (SAMPLE):**
- **Physical Examination:**
  - Utilise a systematic method to assess respiratory distress, including: inspecting the chest, palpating the chest, auscultation, percussion, assessment of any peripheral pulmonary oedema, and assessment of any systemic and cutaneous symptoms.

### 2.3 Vital Signs and Monitoring

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature and blood glucose.
- Undertake spirometry if available for Asthma or COPD.

### 3. Management

#### 3.1 EMT-Level Management

- **Asthma**
  - Ensure the airway is clear and the patient is in a position of comfort.
  - If wheeze present administer salbutamol
  - Assist with the patient's prescribed salbutamol inhaler if available for mild wheeze.
  - Monitor respiratory rate, effort, and SpO<sub>2</sub>.
  - Transport promptly for further assessment.
- **Chronic Obstructive Pulmonary Disease (COPD)**
  - Position the patient upright.
  - Administer oxygen if SpO<sub>2</sub> < 88%, target range as per care plan (typically 88–92%).
  - Administer salbutamol and ipratropium bromide as per DPT.
  - Monitor respiratory status and signs of fatigue.
  - Transport with continuous observation.
- **Anaphylaxis**
  - Identify signs: airway swelling, wheeze, hypotension, urticaria.
  - Administer IM adrenaline 0.5 mg (0.01 mg/kg in children).
  - Administer high-flow oxygen.
  - Lay the patient flat (unless breathing is compromised).
  - Monitor for worsening symptoms.
  - Transport urgently with early pre-alert.

- **Acute Cardiogenic Pulmonary Oedema**
  - Sit patient upright.
  - Administer high-flow oxygen if  $\text{SpO}_2 < 94\%$ .
  - Monitor closely for signs of shock or respiratory distress.
  - Avoid fluid administration.
  - Transport promptly to an appropriate facility.
- **Anxiety**
  - Provide reassurance and a calm environment.
  - Encourage controlled breathing.
  - Assess for physical causes of symptoms (e.g., hypoxia, hypoglycaemia).
  - Avoid over-intervention if symptoms are clearly anxiety-related.
  - Transport if symptoms persist or the diagnosis is unclear.
- **Tuberculosis (TB)**
  - Use PPE, including a mask and gloves, if active TB is suspected.
  - Minimise aerosol-generating procedures.
  - Keep windows open if safe, or isolate within an ambulance.
  - Take detailed history (travel, night sweats, weight loss).
  - Transport with advance notice to the hospital.
- **Sepsis**
  - Recognise red flag signs: altered mental status, tachypnoea, hypotension, fever, rash.
  - Administer high-concentration oxygen.
  - Lay the patient flat if hypotensive.
  - Minimise time on scene.
  - Pre-alert the receiving facility of suspected sepsis.

### 3.2 Paramedic-Level Management

- **Asthma**
  - **EMT-Level Management +**
  - Administer nebulised salbutamol (2.5–5 mg), repeat as needed.
  - Administer ipratropium bromide 500 mcg via nebuliser.
  - Consider IM adrenaline 0.5 mg for life-threatening asthma.
  - Administer corticosteroids (e.g., oral prednisolone or IV hydrocortisone).
  - Support ventilation if required.
- **Chronic Obstructive Pulmonary Disease (COPD)**
  - **EMT-Level Management +**
  - Administer salbutamol and ipratropium via nebuliser.
  - Titrate oxygen to maintain SpO<sub>2</sub> 88–92%.
  - Administer corticosteroids if indicated.
  - Monitor for CO<sub>2</sub> retention and signs of deterioration.
  - Consider assisted ventilation if respiratory failure.
- **Anaphylaxis**
  - **EMT-Level Management +**
  - Repeat IM adrenaline every 5 minutes if no improvement.
  - Establish IV access and administer IV fluids for hypotension.
  - Administer antihistamines (chlorphenamine) and corticosteroids (hydrocortisone).
  - Administer salbutamol via nebuliser if the wheeze persists.
  - Prepare for airway support or intubation if required.

- **Acute Cardiogenic Pulmonary Oedema**
  - **EMT-Level Management +**
  - Administer GTN if systolic BP >100 mmHg.
  - Consider furosemide.
  - Establish IV access.
  - Initiate CPAP if available and trained.
  - Monitor ECG for arrhythmias or signs of ischaemia.
- **Anxiety**
  - **EMT-Level Management +**
  - Rule out organic causes (e.g., sepsis, hypoxia).
  - If appropriate, consider small dose benzodiazepine (e.g., diazepam) in severe agitation.
  - Ensure ongoing monitoring and reassessment.
  - Coordinate with mental health support or crisis teams where appropriate.
- **Tuberculosis (TB)**
  - **EMT-Level Management +**
  - Consider differentials such as pneumonia, lung cancer, or COPD.
  - Administer oxygen via filtered mask if required.
  - Use closed suction systems if available.
  - Notify the hospital of airborne precautions and isolate appropriately.
- **Sepsis**
  - **EMT-Level Management +**
  - Establish IV access.
  - Administer IV fluid bolus (e.g., 20 ml/kg crystalloid).
  - Repeat fluid bolus if hypotension persists.
  - Monitor for signs of fluid overload.
  - Prepare for airway management in deteriorating patients.

#### 4. Key Points

- Respiratory distress can have a clear pathology or present as an undifferentiated disease; when no clear pathology is present, supportive care with transport to an emergency department is the mainstay of treatment.

**M5****ACUTE CORONARY SYNDROME (ACS)**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for assessing and managing patients suspected of having acute coronary syndrome in the prehospital setting.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing patients with suspected acute coronary syndrome.

#### 1.2 Definitions

- **Acute Coronary Syndrome (ACS):** A term used to describe a spectrum of conditions caused by sudden, reduced blood flow to the heart muscle. These include Unstable Angina, Non-ST Segment Elevation MI (NSTEMI) and ST-Segment Elevation MI (STEMI)

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure the safety of providers and the patient.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation and provide oxygen as needed.
  - **Circulation:** Check pulse and perfusion; address signs of shock.

### 2.2 Focused Assessment

- Assess for suspicious factors such as pain radiation, associated symptoms (dyspnoea, hypotension, bradycardia, diaphoresis, or agitation)
- Assess for risk factors and co-morbidities.

### 2.3 Vital Signs and Monitoring

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature and blood glucose.
- Complete 12-lead ECG if available
- Continuously monitor ECG for arrhythmias if available.



### 3. Management

#### 3.1 EMT-Level Management

- Maintain airway, breathing, and circulation.
- Provide oxygen if  $\text{SpO}_2 < 90\%$  or if the patient is short of breath.
- Monitor for worsening signs of poor perfusion.
- Administer Aspirin 300mg to chew if no contraindications
- Transport promptly for further evaluation at the nearest emergency department.
- If the patient is unstable, undertake emergency transfer and notify the receiving facility.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- 12 Lead ECG (if available)
- Administer 400–600 micrograms of GTN sublingually (1 spray or tablet), repeat every 5 minutes up to 3 doses as needed for chest pain, only if systolic BP is  $>90$  mmHg, withhold if hypotensive, tachycardic, bradycardic, or recent use of erectile dysfunction medication.
- If STEMI or New LBBB, transport to a facility capable of thrombolysis or primary percutaneous coronary intervention, and there is a means for the patient to receive treatment.

### 4. Key Points

- Most chest pain is not ACS; have a low threshold for treating a patient with chest pain as ACS, especially if you do not have a 12-lead ECG available. Consider an alternate diagnosis for chest pain.
- Pre-notify the receiving facility in case of STEMI or an unstable patient.

## M6

## BRADYCARDIA AND TACHYCARDIA

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for assessing and managing patients with bradycardia and tachycardia in the prehospital setting.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing patients with bradycardia or tachycardia.

#### 1.2 Definitions

- **Bradycardia:** A heart rate <60 beats per minute (bpm), which may or may not be symptomatic.
- **Tachycardia:** A heart rate >100 bpm, which may or may not be symptomatic.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure the safety of providers and the patient.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions.
  - **Breathing:** Assess adequacy of ventilation and provide oxygen as needed.
  - **Circulation:** Check pulse and perfusion; address signs of shock.

### 2.2 Focused Assessment

#### Bradycardia

- Check for the presence of a pulse and obtain a Rhythm Strip/12-lead ECG if available.
- Assess for:
  - Signs of poor perfusion (e.g., hypotension, altered mental status, chest pain, syncope).
  - Possible causes (e.g., hypoxia, medications, heart block, electrolyte imbalances).

#### Tachycardia

- Check for the presence of a pulse and obtain a Rhythm Strip/12-lead ECG if available.
- Assess for:
  - Hemodynamic stability (e.g., blood pressure, mental status).
  - Symptoms (e.g., chest pain, dyspnoea, palpitations, syncope).
  - Identify the rhythm (e.g., narrow-complex vs. wide-complex tachycardia) using a Rhythm Strip/12-lead ECG.

### 2.3 Vital Signs and Monitoring

- Obtain baseline vitals, including blood pressure, heart rate, respiratory rate, SpO<sub>2</sub>, and blood glucose.
- Continuously monitor ECG for arrhythmias.

### 3. Management

#### 3.1 EMT-Level Management

##### Bradycardia

- Maintain airway, breathing, and circulation.
- Provide oxygen if  $\text{SpO}_2 < 94\%$ .
- Monitor for worsening signs of poor perfusion.
- Rhythm Strip/12-lead ECG if available
- Transport promptly for further evaluation.

##### Tachycardia

- Maintain airway, breathing, and circulation.
- Provide oxygen if  $\text{SpO}_2 < 94\%$ .
- Monitor for hemodynamic stability and prepare for advanced care if needed.
- Rhythm Strip/12-lead ECG if available
- Transport promptly for further evaluation.

#### 3.2 Paramedic-Level Management

##### Bradycardia

- **EMT-Level Management +**
- **Advanced Interventions:**
  - Atropine: 1 mg IV every 3-5 minutes (max dose: 3 mg).
  - If atropine is ineffective:
    - Transcutaneous pacing.
- Address underlying causes (e.g., hypoxia, medications).

## Tachycardia

- **EMT-Level Management +**
- **Narrow-Complex Tachycardia (Stable):**
  - Vagal manoeuvres.
  - Adenosine: 6 mg rapid IV push; may repeat with 12 mg if ineffective.
- **Wide-Complex Tachycardia (Stable):**
  - Consider amiodarone: 150 mg IV over 10 minutes.
- **Unstable Tachycardia:**
  - Immediate synchronized cardioversion.

## 4. Key Points

- Time is critical in bradycardia and tachycardia management; prioritise rapid recognition and intervention.
- Rule out reversible causes before confirming diagnoses.
- Pre-notify the receiving facility to ensure timely care.
- **Paediatric Patients**
  - Bradycardia in children is often secondary to hypoxia; prioritise oxygenation.
  - Use age-appropriate vital sign norms to evaluate tachycardia.
- **Elderly Patients**
  - Symptoms of bradycardia or tachycardia may be atypical.
  - Polypharmacy can contribute to arrhythmias.

**M7****VOMITING**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for recognising and managing vomiting in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing vomiting in prehospital settings.

#### 1.2 Definition

This protocol defines vomiting or emesis as forceful expulsion of stomach contents through the oral cavity, commonly due to several causes including infection, toxins, neurological emergencies and gastrointestinal pathology.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure safety for providers and patients.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation.
  - **Circulation:** Check pulse and perfusion.

### 2.2 Focused Assessment

- **History (SAMPLE):**
  - Signs and symptoms: Frequency, onset, and appearance of vomitus (e.g., blood, bile).
  - Allergies: Assess for allergies to medications.
  - Medications: Identify current medications.
  - Past medical history: Note relevant conditions (e.g., gastrointestinal disorders, pregnancy).
  - Last oral intake: Time and type of last food or drink.
  - Events leading to vomiting: Identify possible triggers.
- **Physical Examination:**
  - Inspect for dehydration (dry mucous membranes, poor skin turgor).
  - Observe for abdominal distension, tenderness, or guarding.
  - Assess for signs of malnutrition.
  - Monitor for neurological signs (e.g., altered mental status).

### 2.3 Vital Signs and Monitoring

- Record SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

- **Airway Protection**
  - Position the patient in the recovery position to reduce aspiration risk if vomiting continues and the patient's airway is unprotected.
  - Suction the airway as necessary.
- **Oxygen Therapy**
  - Administer supplemental oxygen if hypoxia is present.
- **Fluid Resuscitation**
  - Oral Rehydration:
    - Encourage small sips of oral rehydration solution if the patient is alert and tolerating oral intake.
  - Intravenous (IV) Access:
    - Establish IV access for patients with signs of dehydration, hypotension, or persistent vomiting.
    - Administer isotonic crystalloids (e.g., normal saline or lactated Ringer's) as needed if hypotensive.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- IV/IM/PO antiemetic if indicated.



### 3.3 Special Considerations

- **Trauma-Related Vomiting:**
  - Maintain spinal precautions if spinal injury is suspected.
- **Suspected Gastrointestinal Bleeding:**
  - Monitor for signs of shock (e.g., hypotension, tachycardia).
  - Avoid oral intake.
- **Hyperemesis Gravidarum:**
  - Use IV fluids and antiemetics as appropriate.
  - Transport to a facility capable of obstetric care.

### 4. Key Points

- Vomiting can be caused by a wide array of pathologies and is normally unable to be determined prehospital; focus on symptom control and treat any underlying causes as per relevant protocols.

**M8****DIARRHOEA**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for recognising and managing diarrhoea in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing diarrhoea in the prehospital setting.

#### 1.2 Definition

A condition in which faeces are discharged from the bowels frequently and in a liquid form.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure safety for providers and patients.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation.
  - **Circulation:** Check pulse and perfusion.

### 2.2 Focused Assessment

- **History (SAMPLE):**
  - Signs and symptoms: Frequency, onset, and appearance of diarrhoea (e.g., blood, bile, clear).
  - Allergies: Assess for allergies to medications.
  - Medications: Identify current medications.
  - Past medical history: Note relevant conditions (e.g., gastrointestinal disorders, pregnancy).
  - Last oral intake: Time and type of last food or drink.
  - Events leading to diarrhoea: Identify possible triggers.
- **Physical Examination:**
  - Inspect for dehydration (dry mucous membranes, poor skin turgor).
  - Observe for abdominal distension, tenderness, or guarding.
  - Assess for signs of malnutrition.
  - Monitor for neurological signs (e.g., altered mental status).

## 2.3 Vital Signs and Monitoring

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature and blood glucose.

## 3. Management

### 3.1 EMT-Level Management

- **Oxygen Therapy**
  - Administer supplemental oxygen if hypoxia is present.
- **Fluid Resuscitation**
  - Oral Rehydration:
    - Encourage small sips of oral rehydration solution if the patient is alert and tolerating oral intake.
  - Intravenous (IV) Access:
    - Establish IV access for patients with signs of dehydration, hypotension, or persistent vomiting, treat as per relevant protocol.
    - Administer isotonic crystalloids (e.g., normal saline or lactated Ringer's) as needed if hypotensive.

### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Analgesia if pain is present.

## 4. Key Points

- Diarrhoea can be a serious, life-threatening complication of other pathologies in patients with underlying illness, severe malnutrition or comorbidities. Most patients with diarrhoea should be assessed by a physician.
- Do not give any medication to try to stop the diarrhoea.

**M9****DEHYDRATION**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for assessing and managing patients presenting with dehydration in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

This protocol is applicable to EMTs and Paramedics attending to patients of all ages experiencing dehydration, regardless of aetiology.

#### 1.2 Definition

Dehydration is a state in which the body loses more fluids than it takes in, leading to a deficit in total body water and impaired physiological function. It may result from excessive fluid loss (e.g., vomiting, diarrhoea, sweating), inadequate fluid intake, or a combination of both, and can range from mild to life-threatening severity.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure safety for providers and patients. Assess the patient's environment for potential contributing factors (e.g., heat exposure, lack of access to fluids).
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation.
  - **Circulation:** Check pulse and perfusion.

### 2.2 Focused Assessment

- **History (SAMPLE):**
  - Signs and symptoms: Duration, severity, and potential causes of dehydration.
  - Allergies: Note any relevant allergies.
  - Medications: Identify medications that could contribute to dehydration (e.g., diuretics).
  - Past medical history: Note conditions like diabetes, kidney disease, or recent infections.
  - Last oral intake: Assess fluid and food intake over the past 24–48 hours.
  - Events leading to dehydration: Identify any contributing factors (e.g., vomiting, diarrhoea, excessive sweating).
- **Physical Examination:**
  - Look for signs of dehydration:
    - Dry mucous membranes
    - Poor skin turgor
    - Sunken eyes
    - Delayed capillary refill (>2 seconds)
    - Tachycardia or hypotension
    - Altered mental status (e.g., confusion, lethargy)

## 2.3 Vital Signs and Monitoring

- Record SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature and blood glucose.

## 3. Management

### 3.1 EMT-Level Management

- Ensure the airway is patent and breathing is supported.
- Administer supplemental oxygen if hypoxia is present.
- Encourage oral intake of fluids if the patient is alert and tolerating oral intake.
- Use oral rehydration solutions (ORS) when available; avoid plain water in moderate to severe dehydration.
- Monitor vital signs and mental status.
- Transport to the nearest emergency department for further evaluation.
- Monitor for signs of deterioration or complications.

### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Establish IV access for patients with signs of moderate to severe dehydration or haemodynamic instability.
- Administer isotonic crystalloids (e.g., normal saline or lactated Ringer's):
  - **Adults:** 20 ml/kg bolus, reassess after each bolus
  - **Paediatrics:** 10–20 ml/kg bolus, reassess after each bolus
- Adjust fluid administration based on clinical response and patient-specific factors (e.g., age, comorbidities such as renal or cardiac conditions).
- Monitor for complications such as pulmonary oedema.
- Consider glucose level measurement if indicated.
- Adapt care for special populations:
  - **Paediatrics:** Use weight-based fluid calculations; monitor capillary refill, skin turgor, and mental status.
  - **Elderly:** Monitor closely for fluid overload.
  - **Pregnant patients:** Administer IV fluids and transport to a facility capable of obstetric care.

#### 4. Key Points

- Avoid fluid overload with IV rehydration
- Prioritise oral rehydration where possible.



**M10****SEPSIS**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for recognising and managing sepsis in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing sepsis in the prehospital setting.

#### 1.2 Definition

A life-threatening organ dysfunction caused by a dysregulated host response to infection. It is identified clinically by the presence of suspected or confirmed infection along with signs of systemic compromise, such as altered mental status, hypotension, tachycardia, pyrexia, or tachypnoea. Treat patients as having sepsis if they have a suspected or confirmed source of infection with 2 or more of the following:

- Temperature above 38.0oC or below 35.9oC
- Heart rate above 90 bpm
- Respiratory rate above 24 bpm
- Not alert
- Blood Pressure below 90 mmHg
- SPO2 < 94%

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure safety for providers and patients.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation.
  - **Circulation:** Check pulse and perfusion.

### 2.2 Focused Assessment

- **History (SAMPLE):**
- **Physical Examination:**
  - Inspect for the source of infection.
  - Monitor for neurological signs (e.g., altered mental status).

### 2.3 Vital Signs and Monitoring

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

- **Oxygen Therapy**
  - Administer supplemental oxygen if  $SpO_2 < 94\%$
- **Fluid Resuscitation**
  - Rapid transport to the closest emergency department.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Obtain IV access and administer 250 ml of sodium chloride
- IV paracetamol if unable to take oral paracetamol.

### 4. Key Points

- Sepsis is associated with high mortality and morbidity; rapid transport to the emergency department is essential with pre-alert.
- In paediatric patients, suspect sepsis if any of the above criteria are met + mottled skin, reduced feeding, severe malnutrition, diarrhoea or less alert than normal. Administer high-flow oxygen and undertake rapid transfer to the emergency department.

**M11****STROKE / CEREBROVASCULAR ACCIDENT**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for assessing and managing patients with suspected stroke using the BEFAST approach in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing patients with suspected stroke, focusing on rapid identification and transport to appropriate care facilities.

#### 1.2 Definitions

- **Stroke:** A medical emergency caused by the interruption of blood flow to the brain (ischemic) or bleeding in the brain (haemorrhagic).

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure the safety of providers and the patient.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions.
  - **Breathing:** Assess adequacy of ventilation and provide oxygen as needed.
  - **Circulation:** Check pulse and perfusion; address signs of shock.

### 2.2 Focused Assessment

- **BEFAST Assessment:**
  - **Balance:** Observe for unsteadiness or dizziness.
  - **Eyes:** Ask about sudden visual changes or blindness.
  - **Face:** Have the patient smile or show teeth; check for asymmetry.
  - **Arms:** Ask the patient to raise both arms; note weakness or drift on one side.
  - **Speech:** Check for slurred or nonsensical speech; ask the patient to repeat a simple phrase.
  - **Time:** Determine the last known well (LKW) time when the patient was symptom-free.

### 2.3 Vital Signs and Monitoring

- Record SpO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature and blood glucose to rule out hypoglycaemia as a mimic of stroke.

### 3. Management

#### 3.1 EMT-Level Management

##### Basic Supportive Care

- **Positioning:**
  - If conscious: Position patient in a position of comfort, no evidence suggests that supine versus upright has any benefit.
  - If unconscious: Position laterally to protect the airway or consider airway adjuncts.
- **Oxygen Therapy:**
  - Administer oxygen if  $SpO_2 < 90\%$  or signs of hypoxia are present.
- **Transport Priority:**
  - Rapid transport to the nearest facility with a CT scanner if the time from symptom onset to arrival at the hospital will be less than 3 hours. Otherwise, transport to the closest emergency department if no CT scanner is available.
  - Notify the receiving facility of a possible stroke to activate stroke protocols.

#### 3.2 Paramedic-Level Management

##### EMT-Level Management +

##### Advanced Support

- **IV Access:**
  - Establish a large-bore IV for fluid administration if needed.
  - Avoid glucose-containing fluids unless hypoglycaemia is confirmed.
- **Advanced Monitoring:**
  - Continuously monitor ECG for arrhythmias (e.g., atrial fibrillation).
  - Monitor for signs of increased intracranial pressure (e.g., bradycardia, hypertension, altered breathing).

## Thrombolytic Eligibility

- **Gather critical information:**
  - Last Known Well (LKW) time.
  - Current medications, especially anticoagulants.
  - Recent surgeries or trauma.

## 3.3 Special Considerations

### Stroke Mimics

- Rule out conditions that may present similarly to stroke:
  - Hypoglycaemia or hyperglycaemia.
  - Seizures (postictal state).
  - Migraines with aura.
  - Intoxication or drug overdose.

### Paediatric Stroke

- Stroke is rare in children but can occur. Use the same BEFAST assessment and prioritise transport.

### Haemorrhagic Stroke

- Avoid excessive IV fluids that may worsen cerebral oedema.
- Monitor for rapid neurological deterioration.

## 4. Key Points

- Time is critical; prioritise rapid recognition and transport.
- Perform a thorough BEFAST assessment and document findings clearly.
- Rule out stroke mimics, especially hypoglycaemia, before confirming stroke.
- Pre-notify the receiving hospital.

## M12

## SEIZURE

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for recognising and managing seizures in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing patients with suspected or confirmed seizure activity.

#### 1.2 Definitions

- **Seizure:** A sudden, uncontrolled electrical disturbance in the brain that can cause changes in behaviour, movements, sensations, or consciousness.
- **Status Epilepticus:** A seizure lasting longer than 5 minutes or 2 or more discrete seizures without full recovery between them.



## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:**
  - Protect yourself and the patient from hazards.
  - Remove objects that may cause harm during convulsions.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation; provide oxygen as needed.
  - **Circulation:** Check pulse and perfusion.

### 2.2 Focused Assessment

- **Obtain a history of:**
  - Previous seizures or epilepsy diagnosis.
  - Recent head injury, illness, fever, or medication changes.
  - Alcohol or drug use.
  - Duration and characteristics of the seizure (e.g., focal vs. generalised).

### 2.3 Vital Signs and Monitoring

- Record SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

##### During an Active Seizure

- **Protect the patient from injury:**
  - Do not restrain movements.
  - Time the duration of the seizure
  - Place padding under the head if possible.
- Position the patient laterally (recovery position) when convulsions stop to maintain the airway.
- Administer supplemental oxygen via non-rebreather mask if  $\text{SpO}_2 < 94\%$  or signs of respiratory distress.
- If the seizure lasts more than 5 minutes or 2 or more seizures without full recovery between them, consider medical direction to administer midazolam or diazepam IM/Buccal/PR to terminate the seizure if available and initiate rapid transport to the closest emergency department.

##### Postictal Phase (After Seizure Ends)

- Reassure and monitor the patient.
- Assess airway, breathing, and circulation.
- Obtain baseline vitals and monitor mental status.
- Transport priority if:
  - First seizure.
  - Prolonged postictal confusion.
  - Signs of injury or underlying illness.

### 3.2 Paramedic-Level Management

#### During Active Seizure or Status Epilepticus

- **EMT-Level Management +**
- Airway Management:
  - Suction airway as needed.
  - Provide positive pressure ventilation (PPV) if breathing is inadequate.
- Medication Administration (as per local protocol):
  - Midazolam:
    - IM/IV/IN/Buccal: 2-5 mg (adults) or 0.1 mg/kg (paediatrics), repeat as needed up to maximum dose.
  - Diazepam:
    - IV: 5-10 mg (adults) or 0.1-0.2 mg/kg (paediatrics).
  - Lorazepam:
    - IV: 4 mg (adults) or 0.1 mg/kg (paediatrics).

#### Postictal Phase

- **EMT-Level Management +**
- Monitor for recurrent seizures.
- Reassess airway, breathing, and circulation.
- Establish IV/IO access for fluid and medication administration.
- Check blood glucose and correct hypoglycaemia if present.
- Consider underlying causes: Trauma, infection, electrolyte imbalance, or toxicological factors.

### 3.2 Special Considerations

#### Paediatric Seizures

- Consider febrile seizures in children under 5 years with fever.
- Cool the patient gradually if febrile, avoiding rapid cooling methods.

#### Eclampsia (Pregnant Patients)

- Expedite transport to an obstetric-capable facility.
- Treat all pregnant seizures as eclampsia until proven otherwise.

#### Trauma-Related Seizures

- Stabilise the cervical spine if head or neck trauma is suspected.

### 4. Key Points

- Physically protect the patient during seizures without restraining them.
- Airway management is critical to prevent hypoxia and aspiration.
- Address reversible causes (e.g., hypoglycaemia) immediately.
- Early administration of benzodiazepines for status epilepticus improves outcomes.
- Ensure thorough monitoring and reassessment en route to definitive care.

**M13****ALTERED LEVEL OF CONSCIOUSNESS****Purpose**

This protocol provides EMTs and Paramedics with a standardised approach for assessing and managing patients with altered level of consciousness in the prehospital setting.

**1. Scope and Definitions****1.1 Scope**

Applicable to EMTs and Paramedics managing patients with an altered level of consciousness.

**1.2 Definitions**

- **Altered level** of consciousness: Any state where a patient is not fully alert or has a GCS of less than 15. This includes confusion, delirium, reduced GCS and coma.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure the safety of providers and the patient.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation and provide oxygen as needed.
  - **Circulation:** Check pulse and perfusion; address signs of shock.

### 2.2 Focused Assessment

- **Assess pupils.**
- **Common causes include;**
  - Hypoxia
  - Hypoglycaemia
  - Stroke
  - Seizures
  - Trauma (head injury, intracranial haemorrhage)
  - Toxicological causes (overdose, poisoning)
  - Infection (meningitis, sepsis)
  - Metabolic disorders (electrolyte imbalance, hepatic or renal failure)
- A detailed history is imperative using the SAMPLE history. Use bystanders for further information on events prior.

### 2.3 Vital Signs and Monitoring

- Record SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

- Maintain airway, breathing, and circulation.
- Provide oxygen if  $\text{SpO}_2 < 94\%$ .
- Monitor for worsening signs of poor perfusion.
- Correct any abnormal findings such as hypoglycaemia as per the relevant protocol.
- Transport promptly for further evaluation to the nearest emergency department.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Consider advanced airway management if required.

## 4. Key Points

- **Airway Management:**
  - Consider advanced airway interventions (supraglottic airway, endotracheal intubation) if the airway is compromised.
- **Hypoglycaemia:**
  - Administer IV dextrose (e.g., 10% dextrose, 150- 250ml bolus) or IM glucagon if IV access is unavailable.
- **Seizure Management:**
  - If prolonged (>5 minutes) or recurrent seizures, administer benzodiazepines (e.g., midazolam IV/IM/IN as per seizure protocol).
- **Stroke Management:**
  - Rapid transport to a hospital with CT available as per stroke protocols, pre-arrival notification.
  - Consider IV fluids if hypotensive.
- **Toxicological Causes:**
  - Consider naloxone administration for opioid overdose (400 mcg IV/IM/IN, titrated to response).
  - Supportive care for other poisonings or overdoses.
- **Head Trauma:**
  - Manage per traumatic brain injury (TBI) protocols.
  - Maintain adequate cerebral perfusion (avoid hypotension, maintain oxygenation).



**M14****HYPOGLYCAEMIA**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for recognising and managing hypoglycaemia in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing suspected or confirmed hypoglycaemia in patients of all ages.

#### 1.2 Definition of Hypoglycaemia

Hypoglycaemia is typically defined as a blood glucose level below 3.9 mmol/L (70 mg/dL). Symptoms may vary depending on individual thresholds.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure safety for providers and patients.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation.
  - **Circulation:** Check pulse and perfusion.

### 2.2 Focused Assessment

- **Symptoms:** Altered mental status, confusion, diaphoresis, tremors, seizures, or unresponsiveness.
- **History:**
  - Known diabetes or recent insulin/oral use.
  - Recent meal intake, alcohol use, or exercise.

### 2.3 Vital Signs and Monitoring

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

##### Conscious and Able to Swallow

- **Administer oral glucose gel (15-20g):**
  - If glucose gel is unavailable, use other high-sugar substances (e.g., fruit juice, glucose tablets).
  - Reassess blood glucose every 10-15 minutes.

##### Unconscious or Unable to Swallow

- **Positioning:** Place the patient in a recovery position to protect the airway.
- **Rapid transport**

#### 3.2 Paramedic-Level Management

##### Conscious and Able to Swallow

- Follow EMT protocol for oral glucose administration.

##### Unconscious or Unable to Swallow

- **EMT-Level Management +**
- **Establish IV access and administer dextrose:**
  - Adults: 25-50 ml of Dextrose 50% (D50) IV push or 50- 100 ml of Glucose 10%
  - Paediatrics: 0.5-1 g/kg of Dextrose 25% (D25) IV push.
  - Neonates: 0.2-0.5 g/kg of Dextrose 10% (D10) IV push.
- **If IV access is unavailable, administer glucagon:**
  - Adults: 1 mg IM/SC.
  - Paediatrics: 0.5-1 mg IM/SC based on weight.

### Reassessment and Monitoring

- Reassess blood glucose levels every 10-15 minutes.
- Monitor for changes in mental status and vital signs.

### 3.3 Special Considerations

#### Persistent Hypoglycaemia

- Repeat dextrose or glucagon dosing as needed.
- Ensure transport to a facility for definitive care.

#### Complicating Factors

- **Alcohol-related hypoglycaemia:** Higher risk of recurrence; monitor closely.
- **Seizures:** Administer benzodiazepines if indicated per local seizure protocols.

### 4. Key Points

- Rapid recognition of hypoglycaemia is critical.
- Always prioritise airway, breathing, and circulation.
- Treat hypoglycaemia immediately to prevent further complications.
- Reassess frequently and ensure continuous monitoring during transport.

**M15****HYPERGLYCAEMIA**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for recognising and managing hyperglycaemia in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing suspected or confirmed hyperglycaemia in patients of all ages.

#### 1.2 Definition of Hyperglycaemia

Hyperglycaemia is typically defined as a blood glucose level above 11.1 mmol/L (200 mg/dL), with critical levels often exceeding 14 mmol/L (250 mg/dL). Severe hyperglycaemia may lead to diabetic ketoacidosis (DKA) or hyperosmolar hyperglycaemic state (HHS).

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure safety for providers and patients.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation.
  - **Circulation:** Check pulse, perfusion, and hydration status.

### 2.2 Focused Assessment

- **Symptoms:** Polyuria, polydipsia, fatigue, nausea, vomiting, abdominal pain, altered mental status, or fruity odour on breath (in DKA).
- **History:**
  - Known diabetes or recent insulin/oral noncompliance.
  - Recent illness, infection, or stress.

### 2.3 Vital Signs and Monitoring

- Record SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

##### **Mild to Moderate Hyperglycaemia (< 14 mmol/L (250 mg/dL))**

- **Monitor and Support:**
  - Provide reassurance and transport to a healthcare facility or advise on own travel.
  - Encourage hydration with water if the patient is conscious and able to swallow.

##### **Severe Hyperglycaemia or Altered Mental Status**

- **Supportive Care:**
  - Position the patient appropriately to maintain airway patency.
  - Monitor vital signs continuously.
  - Rapid transport to the nearest emergency department

### 3.2 Paramedic-Level Management

#### Severe Hyperglycaemia with Signs of DKA or HHS

- **EMT-Level Management +**
- **IV Access:**
  - Establish IV access and begin fluid resuscitation with **Ringer's lactate or normal saline (0.9%)**:
    - Adults: Administer 250 ml bolus, then reassess.
    - Paediatrics: Administer 10-20 ml/kg over 1 hour, then reassess.
- **Monitor for Complications:**
  - Monitor for fluid overload, especially in patients with cardiac or renal history.
  - Undertake time-critical transfer to the closest emergency department for suspected DKA or altered level of consciousness.
- **Monitoring and Reassessment**
  - Continuously monitor blood glucose levels and mental status.
  - Assess for signs of deterioration, such as increasing respiratory distress, decreasing mental status, or hypotension.

### 3.3 Special Considerations

#### Diabetic Ketoacidosis (DKA)

- **Common in Type 1 diabetes.**
- **Signs:** Fruity odour on breath, deep/laboured breathing (Kussmaul respirations), and abdominal pain.

#### Paediatrics

- Children are at higher risk for cerebral oedema during rapid fluid resuscitation. Administer fluids cautiously.



#### 4. Key Points

- Early recognition of hyperglycaemia prevents complications like DKA and HHS.
- Prioritise airway, breathing, circulation, and hydration.
- Avoid administering insulin in the prehospital setting unless explicitly directed by a physician.
- Reassess frequently and ensure continuous monitoring during transport.

**M16****OVERDOSE / POISONING**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for recognising and managing overdose/poisoning in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing overdose or poisoning in the prehospital setting.

#### 1.2 Definition

The ingestion, inhalation, injection, or absorption of a substance in quantities that overwhelm the body's ability to metabolise or excrete it, leading to harmful physiological effects. This may be accidental or intentional and includes prescription drugs, illicit substances, chemicals, and environmental toxins.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety and Initial Evaluation**
  - Ensure personal and patient safety (e.g., use personal protective equipment, identify hazardous substances).
  - Look for evidence of overdose/poisoning at the scene (e.g., empty pill bottles, chemical containers, drug paraphernalia).
- **Primary Survey (ABCDE)**
  - **Airway:** Assess for patency and protect if compromised.
  - **Breathing:** Check respiratory rate, effort, and oxygen saturation. Note any hypoventilation, apnoea, or signs of respiratory distress.
  - **Circulation:** Assess for pulse, blood pressure, skin signs (e.g., cyanosis, diaphoresis), and capillary refill.

### 2.2 Focused Assessment

- **History Taking (SAMPLE)**
  - **S:** Identify specific signs and symptoms (e.g., altered level of consciousness, seizures).
  - **A:** Ask about known allergies.
  - **M:** Document all medications the patient takes, including recent use.
  - **P:** Explore past medical history, including mental health conditions or substance use disorders.
  - **L:** Last oral intake (substances, food, alcohol).
  - **E:** Events leading to the situation (e.g., intentional ingestion, accidental exposure).

- **Focused Physical Exam**
  - Look for:
    - Track marks or skin abscesses (suggesting IV drug use).
    - Skin signs (e.g., flushing, pallor, diaphoresis, cyanosis).
    - Neurological deficits or altered mental status.
  - Consider toxidromes:
    - **Opioid:** Pinpoint pupils, respiratory depression, altered mental status.
    - **Sympathomimetic:** Dilated pupils, tachycardia, hypertension, agitation, diaphoresis.
    - **Anticholinergic:** Dry skin, flushed face, hyperthermia, dilated pupils, altered mental status.
    - **Cholinergic:** Salivation, lacrimation, urination, diarrhoea, gastrointestinal upset, emesis (SLUDGE symptoms).
- **Recognition of Overdose/Poisoning**
  - Common signs and symptoms include:
    - Altered mental status (e.g., confusion, drowsiness, agitation, unresponsiveness).
    - Abnormal vital signs (e.g., tachycardia, bradycardia, hypotension, hypertension).
    - Respiratory compromise (e.g., hypoxia, bradypnea, hyperventilation).
    - Pupillary changes (e.g., pinpoint or dilated pupils).
    - Vomiting, diarrhoea, abdominal pain.
    - Specific odours (e.g., alcohol, chemicals, burnt almonds for cyanide).

### 2.3 Vital Signs and Monitoring

- Continuously monitor SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

- **Airway and Oxygenation**
  - Ensure a patent airway; consider advanced airway management if necessary.
  - Administer supplemental oxygen to maintain SpO<sub>2</sub> ≥ 94%.
- **Ventilation Support**
  - Provide bag-valve mask (BVM) ventilation for patients with inadequate respiratory effort.
- **Specific Interventions by Suspected Cause**
  - Opioid Overdose:
    - Administer naloxone (Narcan):
      - Adults: 400 mcg IV/IM/IN, repeated every 2-3 minutes as needed (maximum total dose 3 mg).
      - Paediatrics: 0.01 mg/kg IV/IM/IN, repeated as necessary.
    - Be prepared for agitation or withdrawal symptoms.
  - Cholinergic Poisoning (e.g., organophosphates):
    - Administer atropine 2 mg IV/IM, repeated every 5 minutes until secretions improve.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- **Beta-Blocker or Calcium Channel Blocker Overdose:**
  - Administer fluids for hypotension.
  - Consider glucagon if available per protocol.
- **Tricyclic Antidepressant Overdose:**
  - Treat wide QRS complexes or arrhythmias with sodium bicarbonate (1 mEq/kg IV).

#### 4. Key Points

- Remove contaminated clothing and rinse the skin with water for chemical exposures.
- Do not induce vomiting.
- Consider if the overdose was intentional, unintentional, or if it was an intentional poisoning.
- For those patients who have taken an intentional overdose of a toxic substance, they should be assessed by a physician irrespective of clinical assessment findings.

## M17

## SUSPECTED SEVERE MALARIA

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for assessing and managing patients with suspected severe malaria in the prehospital setting.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing patients with fever and systemic signs in malaria endemic areas.

#### 1.2 Definitions

**Severe Malaria:** A life-threatening parasitic infection, most often caused by *Plasmodium falciparum*, that can result in coma, seizures, shock, hypoglycaemia, and multi-organ dysfunction.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure standard infection precautions.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation.
  - **Circulation:** Check pulse and perfusion.

### 2.2 Focused Assessment

- Fever, confusion, vomiting, or seizure activity
- Assess for hypoglycaemia and altered GCS
- Consider differential diagnoses: meningitis, sepsis, febrile convulsions

### 2.3 Vital Signs and Monitoring

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature and blood glucose.
- Monitor for signs of shock or convulsions



### 3. Management

#### 3.1 EMT-Level Management

- Airway management as required
- Oxygen therapy as required
- Manage hypoglycaemia (oral glucose if GCS intact) or IV glucose 10%
- Urgent transport with pre-alert if severe features present

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Establish IV access and administer a 250ml fluid bolus if hypotensive.
- Administer IV glucose if blood glucose <4.0 mmol/L.
- If seizure present, treat as per seizure protocol.

### 4. Key Points

- Severe malaria can mimic other conditions like meningitis and sepsis.
- Hypoglycaemia and seizures are common presentations in children.
- Pre-alert the receiving hospital if suspected severe malaria, especially in patients with reduced consciousness or seizures.

## M18

## SUSPECTED MENINGITIS

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for assessing and managing patients with suspected meningitis in the prehospital setting.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics assessing patients with fever, altered consciousness, neck stiffness, or non-blanching rash.

#### 1.2 Definitions

**Meningitis:** Inflammation of the meninges, often due to viral or bacterial infection. Bacterial meningitis is a medical emergency that may lead to sepsis, seizures, and death.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure the safety of providers and the patient, and ensure adequate PPE is worn.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation and provide oxygen as needed.
  - **Circulation:** Check pulse and perfusion; address signs of shock.

### 2.2 Focused Assessments

- Fever, headache, photophobia, vomiting
- Neck stiffness, non-blanching rash, altered mental state
- Seizure activity or post-ictal state
- Recent infections or sick contacts

### 2.3 Vital Signs and Monitoring

- Record SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature and blood glucose.
- Monitor for signs of sepsis or raised intracranial pressure

### 3. Management

#### 3.1 EMT-Level Management

- Oxygen therapy if  $\text{SpO}_2 < 94\%$  or signs of respiratory distress
- Position appropriately (consider lateral if GCS reduced)
- Apply a surgical mask to the patient if actively coughing

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- IV fluid resuscitation if hypotensive
- If seizure present, treat as per seizure protocol.

### 4. Key Points

- Meningococcal meningitis can present with rapid deterioration and rash.
- Early recognition and transport are critical.
- Notify the receiving facility immediately if meningitis is suspected.

**M19****ACUTE BEHAVIOURAL DISTURBANCE**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for recognising and managing acute behavioural disturbance in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing acute behavioural disturbance in the prehospital setting.

#### 1.2 Definition

**Acute Behavioural Disturbance (ABD)** refers to a state of severe agitation, aggression, and altered mental state that may be associated with serious underlying medical or psychiatric conditions. It often presents with features such as confusion, hallucinations, shouting, violence, increased strength, and a lack of response to verbal commands or pain. It may progress rapidly to sudden collapse and death if not recognised and managed urgently. Often caused by;

- Psychiatric conditions (e.g., psychosis, bipolar disorder, schizophrenia).
- Substance use (e.g., intoxication, withdrawal).
- Medical conditions (e.g., hypoglycaemia, hypoxia, infection, traumatic brain injury).

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety and Initial Approach**
  - Ensure the safety of EMS personnel, the patient, and bystanders.
  - Request additional resources (police) if needed.
  - Use a calm and non-confrontational approach.
  - Maintain situational awareness and establish a safe escape route if necessary.
- **Primary Survey (ABCDE)**
  - **Airway:** Assess for obstruction or compromise.
  - **Breathing:** Check respiratory rate, oxygen saturation, and signs of distress.
  - **Circulation:** Evaluate heart rate, blood pressure, and capillary refill.

## 2.2 Focused Assessment

- **Identify Potential Causes of ABD**
  - Look for medical or traumatic causes:
    - **Hypoxia:** Low oxygen saturation.
    - **Hypoglycaemia:** Altered mental status, diaphoresis.
    - **Head trauma:** Scalp lacerations, bruising, or unequal pupils.
    - **Substance use:** Smell of alcohol, drug paraphernalia, or patient history.
    - **Psychiatric conditions:** Pre-existing mental health diagnosis or history of psychiatric care.
- **Focused History (SAMPLE)**
  - **S:** Document symptoms, including agitation, confusion, hallucinations, or aggression.
  - **A:** Ask about allergies to medications.
  - **M:** Review current and past medications.
  - **P:** Note past medical or psychiatric history, including previous episodes of ABD.
  - **L:** Identify last oral intake or substance use.
  - **E:** Determine events leading to the episode (e.g., stressors, triggers, injuries).

## 2.3 Vital Signs and Monitoring

- Continuously monitor SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

- **De-escalation Techniques**
  - Use verbal de-escalation:
    - Speak in a calm, reassuring tone.
    - Avoid confrontation or threats.
    - Establish rapport and validate the patient's feelings.
  - Reduce environmental stimuli (e.g., dim lights, reduce noise).
  - Maintain a safe distance and avoid physical contact unless necessary.
- **Non-Pharmacological Interventions**
  - Provide reassurance and clear explanations of actions.
  - Encourage cooperation while maintaining patient dignity.
  - Use family members or caregivers for support if appropriate.



### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- **Pharmacological Interventions**
  - Consider sedation when the patient poses an immediate risk to themselves or others and verbal de-escalation fails.
  - Options include:
    - **Midazolam:** 2.5-5 mg IM/IV (adult dose).
    - **Haloperidol/Doperadol:** 5-10 mg IM (use cautiously, monitor for QT prolongation).
    - **Ketamine:** 1-2 mg/kg IV or 4-5 mg/kg IM for severe agitation (requires close monitoring).
  - Monitor for side effects (e.g., respiratory depression, hypotension) and manage as necessary.
- **Physical Restraints**
  - Use only as a last resort when the patient's behaviour poses an imminent threat.
  - Ensure restraints are applied in a safe and humane manner.
  - Monitor the patient closely for respiratory distress, positional asphyxia, or circulatory compromise.
- **Treat Underlying Medical Causes**
  - **Hypoglycaemia:** Administer oral glucose or IV dextrose (e.g., D10 or D50). IM Glucagon if needed.
  - **Hypoxia:** Provide high-flow oxygen to maintain SpO<sub>2</sub> ≥ 94%.
  - **Substance intoxication:** Treat based on specific toxidrome (e.g., naloxone for opioid overdose).

### 4. Key Points

- Transport to an appropriate facility based on the suspected cause (e.g., psychiatric hospital, emergency department).
- Notify the receiving facility of:
  - The patient's condition and behaviour.
  - Medications administered and response to treatment.
  - Any safety concerns.
- Consider involving law enforcement for patient and responder safety during transport if necessary.

## M20

## ACUTE ABDOMEN

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for assessing and managing patients with abdominal pain in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing patients of all ages with acute or chronic abdominal pain.

#### 1.2 Definitions

- **Abdominal Pain:** Pain originating from the abdominal region, which may result from various medical, surgical, or traumatic causes.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure the safety of providers and the patient.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions.
  - **Breathing:** Assess adequacy of ventilation and oxygenation.
  - **Circulation:** Check for signs of shock (e.g., pale, cool, clammy skin; weak pulse).

### 2.2 Focused Assessment

- **History (SAMPLE & OPQRST):**
  - **Signs and Symptoms:** Location, onset, severity, quality, and radiation of pain.
  - **Allergies.**
  - **Medications:** Recent use of pain relievers, laxatives, or anticoagulants.
  - **Past Medical History:** History of gastrointestinal, renal, or gynaecological issues. **Pregnancy?**
  - **Last Oral Intake.**
  - **Events Leading Up to the Pain.**
- **Examination:**
  - Inspect for distension, scars, bruising, or pulsations.
  - Auscultate for bowel sounds.
  - Palpate for tenderness, guarding, or rigidity, including;

**McBurney's Sign**

- Purpose: Indicates appendicitis.
- Assessment: Palpate McBurney's point (1/3 of the distance from the anterior superior iliac spine to the umbilicus on the right side).
- Positive Sign: Marked tenderness at this point suggests acute appendicitis.

**Murphy's Sign**

- Purpose: Assesses for cholecystitis (inflammation of the gallbladder).
- Assessment: Place your hand at the right costal margin in the mid-clavicular line and ask the patient to take a deep breath.
- Positive Sign: Sudden pain and arrest of inspiration due to the inflamed gallbladder pressing against the examiner's fingers.

**Rovsing's Sign**

- Purpose: Also suggests appendicitis.
- Assessment: Apply deep palpation in the left lower quadrant.
- Positive Sign: Pain is felt in the right lower quadrant, indicating peritoneal irritation associated with appendicitis.

**2.3 Vital Signs and Monitoring**

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

##### Basic Supportive Care

- **Positioning:** Position the patient for comfort, typically in a supine or semi-Fowler's position.
- **Oxygen Therapy:**
  - Administer oxygen via nasal cannula or non-rebreather mask if  $\text{SpO}_2 < 94\%$  or signs of hypoxia.
- **Transport Priority:**
  - Expedite transport if the patient exhibits signs of shock, severe pain, or peritonitis.

#### 3.2 Paramedic-Level Management

##### EMT-Level Management +

##### Advanced Support

- **IV Access:**
  - Establish IV access.
  - Administer a fluid bolus for signs of hypovolemia or dehydration:
    - Adults: 500-1,000 ml; titrate to effect.
    - Paediatrics: 10-20 ml/kg.
- **Pain Management (per protocol):**
- **Nausea and Vomiting:**
  - Ondansetron: 4 mg IV/IM/ODT for adults; 0.1 mg/kg for paediatrics.

##### Monitoring and Reassessment

- Continuously monitor vital signs, mental status, and pain levels.
- Assess for worsening abdominal distension, rigidity, or other signs of deterioration.

### 3.3 Special Considerations

#### Differential Diagnosis

- **Acute Abdomen:** Surgical emergencies like perforation or obstruction.
- **Medical Causes:** Gastroenteritis, pancreatitis, diabetic ketoacidosis.
- **Gynaecological Causes:** Ectopic pregnancy, ovarian torsion, or pelvic inflammatory disease.
- **Trauma:** Blunt or penetrating injury to the abdomen.

#### Paediatrics

- Consider intussusception, appendicitis, or gastroenteritis.
- Maintain a high index of suspicion for non-accidental trauma.

#### Elderly Patients

- Symptoms may be atypical, and pain may be less pronounced.
- Common causes include bowel obstruction, ischemic bowel, and abdominal aortic aneurysm.

#### Pregnant Patients

- Consider ectopic pregnancy or complications of pregnancy (e.g., preeclampsia).
- Position in left lateral recumbent position to prevent supine hypotension.

### 4. Key Points

- Abdominal pain can indicate life-threatening conditions; prioritise early recognition and transport.
- Maintain a broad differential diagnosis and reassess frequently.
- Avoid giving oral medications or fluids unless explicitly authorised.
- Provide comfort measures and support en route to definitive care.

**M21****SICKLE CELL CRISIS**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for assessing and managing patients experiencing a sickle cell crisis in the prehospital setting.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing patients known or suspected to have sickle cell disease presenting with pain, respiratory symptoms, or signs of systemic illness.

#### 1.2 Definitions

- **Sickle Cell Disease (SCD):** A genetic condition resulting in the production of abnormal haemoglobin (HbS), leading to the formation of rigid, sickle-shaped red blood cells.
- **Sickle Cell Crisis:** A painful and potentially life-threatening episode often triggered by infection, dehydration, cold exposure, or hypoxia, resulting in vaso-occlusion, ischaemia, and tissue damage.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure the safety of providers and the patient.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation and provide oxygen as needed.
  - **Circulation:** Check pulse and perfusion; address signs of shock.

### 2.2 Focused Assessment

- Severe generalised or localised pain (often limbs, chest, back, abdomen)
- Breathlessness or chest pain (may indicate acute chest syndrome)
- History of SCD, previous crises, or hospitalisations
- Look for signs of infection or dehydration.

### 2.3 Vital Signs and Monitoring

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature and blood glucose.
- Monitor for signs of sepsis or hypoxia



### 3. Management

#### 3.1 EMT-Level Management

- High-flow oxygen if  $SpO_2 < 94\%$  or signs of hypoxia
- Position the patient for comfort and reduce exertion.
- Encourage oral fluids if safe and alert.
- Provide reassurance and early pain assessment.
- Prompt transport and pre-alert if chest symptoms or altered level of consciousness are present.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Establish IV access.
- Administer IV fluids if dehydrated or hypotensive.
- Administer analgesia IV/IM NSAID e.g. Diclofenac 75mg IV/SC
- Monitor for acute chest syndrome or signs of sepsis.
- Prepare for airway support if deteriorating

### 4. Key Points

- Pain relief is a priority—do not delay analgesia.
- Acute chest syndrome may mimic pneumonia—treat any respiratory symptoms seriously.
- Many patients with SCD are young and familiar with their condition—listen to their concerns and typical management needs.
- Pre-alert for any signs of respiratory distress, altered consciousness, or suspected infection.

2



Obstetrics

**01****NORMAL CHILDBIRTH**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for safely assessing and managing normal, uncomplicated childbirth in the prehospital setting.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing normal childbirth.

#### 1.2 Definitions

**Normal childbirth:** The spontaneous onset and progression of labour resulting in vaginal delivery at term (37–42 weeks gestation) without the use of instruments and without maternal or neonatal complications.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure the safety of providers and the patient.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation and provide oxygen as needed.
  - **Circulation:** Check pulse and perfusion; address signs of shock.
- Determine gestational age, parity, and duration of labour.
- Assess contraction frequency, urge to push, and perineal bulging or crowning.

### 2.2 Focused Assessment

- Confirm singleton pregnancy.
- Determine if membranes have ruptured.
- Check for abnormal vaginal bleeding or signs of complications (e.g. meconium-stained liquor, malpresentation).
- Consider differential diagnoses (e.g. preterm labour, placental abruption).

### 2.3 Vital Signs and Monitoring

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

##### Delivery Process

- Support the perineum as the head crowns – apply gentle pressure to control expulsion.
- Guide the delivery of the head – check for nuchal cord and gently unwrap if present.
- Allow spontaneous restitution and delivery of the shoulders.
- Deliver the body and legs with minimal handling.

##### Immediate Post-Delivery Care

- Dry and stimulate the baby to initiate respiration.
- Assess APGAR score at 1 and 5 minutes.
- Clamp and cut the cord after pulsations cease (typically after 1–3 minutes).
- Keep mother and baby warm – encourage skin-to-skin contact.
- Monitor for signs of postpartum haemorrhage and manage as needed.
- Deliver the placenta – do not pull; allow spontaneous expulsion if time permits.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Inhaled analgesia, where available, and if required

### 4. Key Points

- If delivery is imminent, remain on scene and deliver safely.
- If early labour or concerns arise, transport to the nearest maternity unit.
- Maintain maternal dignity and provide reassurance throughout.
- Do not attempt to delay delivery if crowning is observed.
- Ensure neonatal and maternal vital signs are monitored continuously post-delivery.
- Prepare for potential complications such as PPH or neonatal resuscitation.

## 02

## BREECH CHILDBIRTH

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for recognising and managing breech birth presentations in the prehospital setting.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing breech birth.

#### 1.2 Definitions

**Breech Birth:** A type of childbirth where the baby is delivered buttocks or feet first instead of the normal head-first position. Types include frank, complete, and footling breech.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure the safety of providers and the patient.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation and provide oxygen as needed.
  - **Circulation:** Check pulse and perfusion; address signs of shock.
- Determine gestational age and assess for signs of imminent delivery.
- Identify breech presentation: visible or palpable buttocks or feet at the perineum.

### 2.2 Focused Assessment

- Check for signs of complications (e.g. cord prolapse, delayed descent).
- Monitor for abnormal bleeding, prolonged second stage, or foetal distress.
- Consider time since rupture of membranes.

### 2.3 Vital Signs and Monitoring

- Record SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

- Do not attempt to reposition or delay breech delivery once birth is imminent.
- Allow spontaneous delivery of the buttocks and legs.
- Support the baby's body as it delivers, ensuring gentle downward traction.
- Keep the baby's body warm and supported.

#### **Delivery of the Head (Mauriceau-Smellie-Veit manoeuvre)**

- Place the baby astride the forearm, with the middle finger on the maxilla (not in the mouth) to maintain flexion.
- Apply gentle downward traction with the other hand supporting the shoulders.
- Elevate the baby's body once the hairline is visible to deliver the head.
- If the head does not deliver, consider suprapubic pressure or rotate the baby 90° to release the arms.

#### **Complications**

- Be prepared for neonatal resuscitation.
- Do not pull forcefully – allow for gentle traction only.
- If the head fails to deliver within 3 minutes of the body, initiate emergency transfer.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Inhaled analgesia if available and required.



#### 4. Key Points

- If breech presentation is identified but delivery is not imminent, transport urgently to a maternity unit.
- If delivery is unavoidable, proceed cautiously and notify the receiving facility immediately.
- Breech birth can proceed safely with minimal intervention if delivery is spontaneous.
- Do not delay delivery once the body has been delivered – assist with head only if necessary.
- Avoid unnecessary handling or excessive traction.

## 03

## SHOULDER DYSTOCIA

**Purpose**

This protocol provides EMTs and Paramedics with a standardised approach for recognising and managing shoulder dystocia during childbirth in the prehospital setting.

**1. Scope and Definitions****1.1 Scope**

Applicable to EMTs and Paramedics managing shoulder dystocia.

**1.2 Definitions**

**Shoulder Dystocia:** A delivery complication in which the baby's anterior shoulder becomes impacted behind the maternal pubic symphysis after the head has been delivered, preventing further progress of birth.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure the safety of providers and the patient.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation and provide oxygen as needed.
  - **Circulation:** Check pulse and perfusion; address signs of shock.

#### Initial Recognition

- Head delivers but retracts against the perineum (“turtle sign”).
- Failure of restitution or descent of the shoulders.
- Difficulty with delivering the face or chin.
- Prolonged head-to-body delivery time (>60 seconds).

### 2.2 Focused Assessment

- Consider maternal diabetes, obesity, previous shoulder dystocia, large baby (macrosomia), or post-dates pregnancy.
- Prepare for immediate neonatal resuscitation.

### 2.3 Vital Signs and Monitoring

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

##### Immediate Actions

- Call for additional Paramedic or physician support—shoulder dystocia is a time-critical emergency.
- Do not apply traction to the baby's head.

##### McRoberts' Manoeuvre

- Position the mother supine with knees drawn to chest.
- Hyperflex the hips – this may free the impacted shoulder.
- Apply suprapubic pressure (not fundal pressure) – use firm downward pressure above the pubic bone to dislodge the anterior shoulder; if this is unsuccessful, apply intermittent pressure.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- If trained and competent, attempt internal rotation or delivery of the posterior arm.
- Proceed rapidly to emergency transport if initial manoeuvres fail.

### 4. Key Points

- Prepare for neonatal resuscitation.
- If shoulder dystocia is resolved and delivery is completed, continue to the maternity unit for postnatal monitoring.
- **Shoulder dystocia is time-critical – aim to resolve shoulder dystocia within 5 minutes of head delivery or initiate emergency transfer to the closest emergency department.**
- Use McRoberts' and suprapubic pressure as first-line interventions.
- Never apply excessive force to the baby's head or neck.

**04****CORD PROLAPSE**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for identifying and managing umbilical cord prolapse during labour in the prehospital setting.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing umbilical cord prolapse.

#### 1.2 Definitions

**Cord Prolapse:** A condition in which the umbilical cord descends through the cervix ahead of or alongside the presenting part of the foetus, risking cord compression and compromised foetal oxygenation.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure the safety of providers and the patient.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation and provide oxygen as needed.
  - **Circulation:** Check pulse and perfusion; address signs of shock.

### 2.2 Focused Assessment

- Initial Recognition
  - Visible or palpable cord in the vagina or at the perineum.
  - Sudden deceleration in foetal heart rate (if monitored) or reduced foetal movements.
  - May occur following rupture of membranes, especially in high-risk pregnancies.
- Risk Factors
  - Prematurity, polyhydramnios, malpresentation (e.g. breech, transverse lie), multiple pregnancy, or long umbilical cord.

### 2.3 Vital Signs and Monitoring

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

- Call for urgent backup and prepare for emergency transport.
- Instruct the mother to adopt the knee-chest position or exaggerated Sims' position (left lateral with pillow under hip) to relieve pressure on the cord.
- Do not attempt to push the cord back into the vagina.
- Keep exposed cord warm and moist with sterile saline-soaked gauze.

#### Ongoing Care

- Reassure and monitor the mother continuously.
- Avoid unnecessary handling or movement that may increase cord compression or vasospasm.
- Prepare for possible neonatal resuscitation upon delivery.

#### 3.2 Paramedic-Level Management

- EMT-Level Management +
- Inhaled analgesia if available and required

### 4. Key Points

- Initiate time-critical transfer to the nearest emergency department.
- Cord prolapse is an obstetric emergency with a high risk of foetal hypoxia.
- Prioritise relieving cord pressure and rapid transport.
- Maintain warmth and moisture of the cord if exposed.

## 05

## PRE ECLAMPSIA / ECLAMPSIA

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for recognising and managing pre-eclampsia and eclampsia in the prehospital setting.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing patients with pre-eclampsia or eclampsia.

#### 1.2 Definitions

**Pre-Eclampsia:** A condition typically occurring after 20 weeks of gestation, characterised by new-onset hypertension ( $\geq 140/90$  mmHg) and proteinuria or signs of organ dysfunction (e.g. headache, visual disturbance, oedema).

**Eclampsia:** The onset of seizures in a patient with pre-eclampsia, unrelated to other neurological causes.



## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure the safety of providers and the patient.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation and provide oxygen as needed.
  - **Circulation:** Check pulse and perfusion; address signs of shock.

### 2.2 Focused Assessment

#### Initial Recognition

- High blood pressure ( $\geq 140/90$  mmHg).
- Persistent headache, visual disturbances (e.g. flashing lights), or upper abdominal pain.
- Facial or hand oedema, brisk reflexes, altered mental status.
- Seizure activity (eclampsia).
- **Focused Assessment**
- Confirm gestational age and medical history.
- Assess for signs of organ dysfunction (e.g. reduced consciousness, chest pain, oliguria).
- Monitor maternal vital signs closely.
- If available, assess foetal movements or heart tones.

### 2.3 Vital Signs and Monitoring

- Record SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

- Maintain airway, breathing, and circulation.
- Administer high-flow oxygen if oxygen saturation < 94%.
- Minimise stimulation – dim lights and keep the environment quiet.
- Place the patient in the left lateral position to optimise uteroplacental perfusion.
- If SBP > 140 mmHg or DBP 100 mmHg with any symptoms, undertake emergency transfer to the closest emergency department.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- If seizing, protect the airway and prevent injury during a seizure.
- Administer Magnesium Sulphate 4g IV/IM if trained and authorised.
- If magnesium is unavailable, administer Diazepam 5–10 mg IV or PR or Midazolam 1-2mg IV/10mg Buccal.
- Monitor for signs of magnesium toxicity (loss of reflexes, respiratory depression).
- Undertake emergency transfer to the closest emergency department.

### 4. Key Points

- Eclampsia is life-threatening for both mother and baby – act quickly.
- Magnesium Sulphate is the preferred first-line treatment for seizure prevention and control. Only if Magnesium Sulphate fails or is not available should you follow seizure protocol.
- Minimise sensory stimulation to prevent seizure recurrence.
- Early recognition and rapid transport are key to maternal and foetal survival.

**06****ANTE-PARTUM HAEMORRHAGE**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for recognising and managing antepartum haemorrhage in the prehospital setting.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing antepartum haemorrhage.

#### 1.2 Definitions

Ante-Partum Haemorrhage (APH): Vaginal bleeding occurring after 20 weeks gestation and before the birth of the baby. Common causes include placenta praevia, placental abruption, and vasa praevia.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure the safety of providers and the patient.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation and provide oxygen as needed.
  - **Circulation:** Check pulse and perfusion; address signs of shock.

### APH Assessment

- Estimate the amount, colour, and duration of bleeding.
- Determine gestational age and foetal movements.
- Avoid vaginal examination.

### 2.2 Focused Assessment

- Focused Obstetric History
  - Ask about previous APH, placenta praevia, trauma, contractions, or pain.
  - Determine if membranes have ruptured or if there is any associated pain (e.g. with abruption).
  - Assess for risk factors: multiple pregnancies, hypertension, smoking, previous caesarean section.

### 2.3 Vital Signs and Monitoring

- Record SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

##### General Management

- Position the patient in the left lateral position.
- Administer high-flow oxygen if saturation <94% or signs of hypoperfusion are present.
- Monitor maternal vital signs and observe for deterioration.
- Do not perform any vaginal or pelvic examinations.
- If available, monitor foetal heart tones or assess foetal movements.

##### Bleeding Control and Fluids

- Pad and assess ongoing blood loss visually, document findings.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Initiate IV access and consider fluid resuscitation with crystalloid solution if signs of shock are present
- Consult medical direction for Tranexamic Acid administration authorisation if the mother is peri-arrest with prolonged transport time.

### 4. Key Points

- Do not perform vaginal examinations in any patient with APH.
- Treat all APH as potentially life-threatening to both mother and foetus.
- Monitor closely for signs of shock – compensate early.
- Undertake time-critical transfer to the closest emergency department.

**07****POST-PARTUM HAEMORRHAGE**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for recognising and managing post-partum haemorrhage (PPH) in the prehospital setting.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing post-partum haemorrhage.

#### 1.2 Definitions

- Post-Partum Haemorrhage (PPH): Blood loss exceeding 500 mL following vaginal delivery or 1000 mL following caesarean section.
- Primary PPH occurs within the first 24 hours of delivery; secondary PPH occurs after 24 hours and within 6 weeks postpartum.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure the safety of providers and the patient.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation and provide oxygen as needed.
  - **Circulation:** Check pulse and perfusion; address signs of shock.

### 2.2 Focused Assessment

#### PPH Assessment

- Identify signs of major bleeding: soaking of pads/towels, pooling of blood.
- Assess for signs of hypovolaemic shock: tachycardia, hypotension, pallor, dizziness, altered consciousness.
- Ask about delivery time, placental delivery, and any complications during birth.

#### Risk Factors

- Retained placenta, uterine atony, trauma (perineal or uterine), previous PPH, multiple pregnancy, prolonged labour.

### 2.3 Vital Signs and Monitoring

- Record SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

##### Immediate Actions

- Call for urgent Paramedic support and prepare for emergency transport.
- Lie the mother flat and keep warm.
- Administer **high-flow oxygen** if  $\text{SpO}_2 < 94\%$  or signs of shock are present.
- **Massage the uterus** firmly through the abdomen to stimulate contraction (fundal massage).
- If the placenta is not delivered, do not attempt manual removal.

##### Internal Bimanual Compression

- If uterine atony and bleeding persist despite uterotonics and fundal massage:
  - Insert a gloved hand into the vagina and form a fist pressing against the anterior wall of the uterus via the vagina.
  - Simultaneously, the other hand is used externally on the abdomen to compress the uterus between both hands.
  - Maintain compression until bleeding is controlled or handover to hospital staff.

##### Aortic Compression

- If bimanual compression is ineffective and bleeding continues:
  - Use **external aortic compression** to reduce pelvic blood flow:
    - Locate the abdominal aorta (just above the umbilicus, slightly left of midline).
    - Apply firm downward pressure using a closed fist or flat hand to compress the aorta against the spinal column.
  - Confirm by feeling reduced femoral pulses and noting decreased bleeding.



### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Pharmacological Management
  - If authorised and trained, administer Oxytocin or Syntometrine IM, + 1g Tranexamic Acid IV/IM.
  - Gain bilateral large-bore IV access and administer sodium chloride or lactated Ringer's in 250ml boluses to aim for a systolic BP of 90mmHg.

### 4. Key Points

- Initiate time-critical transfer
- Maintain uterine massage, compression techniques, and ongoing observations en route.
- Fundal massage is the first-line intervention in all cases of suspected uterine atony.
- Internal bimanual compression and aortic compression are critical life-saving techniques when bleeding is not controlled.
- Do not delay transfer – initiate treatment en route.
- Maintain warmth, oxygenation, and close monitoring throughout.

3



Paediatrics

**P1****NEONATAL RESUSCITATION**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for assessing and resuscitating newborns immediately after birth.

### 1. Scope and Definitions

#### 1.1 Scope

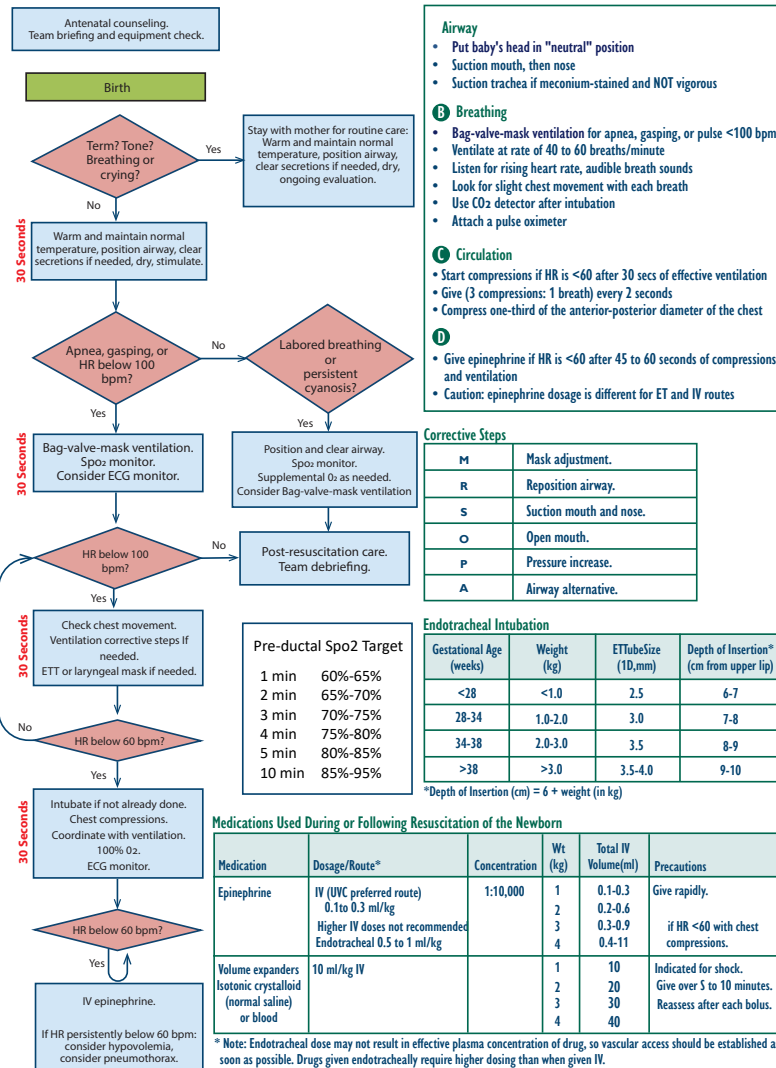
Applicable to EMTs and Paramedics managing neonatal resuscitation.

#### 1.2 Definitions

**Neonatal Resuscitation:** The provision of life-support measures to a newborn who does not initiate or sustain adequate breathing or circulation at birth.

## 2. Assessment

The most important and effective action in neonatal resuscitation is ventilation of the baby's lungs.



## 3. Management

### 3.1 EMT-Level Management

- Follow the algorithm, and provide rapid transport to the emergency department after 5 minutes of ventilation and CPR if there is no response.

### 3.2 Paramedic-Level Management

- EMT-Level Management +**
- Adrenaline as per the algorithm
- ET Intubation if trained and competent as per the algorithm

## 4. Key Points

- Initiate time-critical transfer if no response to resuscitation within 5 minutes of delivery.

**P2****PAEDIATRIC RESPIRATORY DISTRESS**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for assessing and managing paediatric respiratory emergencies, including bronchiolitis, croup, and asthma.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing patients under 16 years old with respiratory distress.

#### 1.2 Definitions

**Paediatric Respiratory Emergency:** Conditions such as bronchiolitis, croup, or asthma presenting with respiratory distress in children.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure the safety of providers and the patient.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation and provide oxygen as needed.
  - **Circulation:** Check pulse and perfusion; address signs of shock.

### 2.2 Focused Assessment

- Stridor (croup), wheeze (asthma), crackles (bronchiolitis)
- Fever, nasal congestion, poor feeding
- Seesaw breathing, head bobbing in infants.

### 2.3 Vital Signs and Monitoring

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.
- Monitor for signs of fatigue or hypoxia.

### 3. Management

#### 3.1 EMT-Level Management

- Position upright or in the parent's arms.
- Oxygen via nasal cannula if tolerated.
- Minimise interventions—keep the child calm.
- Prompt transport

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- PO/IV/IM dexamethasone and nebulised adrenaline for moderate-severe croup
- Nebulised salbutamol/ipratropium bromide for wheeze (>1 year)
- Consider corticosteroids if indicated and available

### 4. Key Points

- Keeping the child calm is a priority—distress worsens breathing.
- Avoid aerosol-generating procedures unless necessary.
- Pre-alert for any child with severe respiratory distress or fatigue.

## P3

## FEBRILE CHILD

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for recognising and managing paediatric patients with fever, including differentiating between common and serious illness.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing febrile children under 16 years of age.

#### 1.2 Definitions

**Febrile Child:** A child with a temperature  $\geq 38^{\circ}\text{C}$ , presenting with fever as a primary symptom.



## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure the safety of providers and the patient.
- **Primary Survey (ABCDE):**
  - **Airway:** Ensure patency and clear obstructions
  - **Breathing:** Assess adequacy of ventilation and provide oxygen as needed.
  - **Circulation:** Check pulse and perfusion; address signs of shock.

### 2.2 Focused Assessment

- Fever, vomiting, seizure history, rash, reduced feeding
- Sick contacts
- AVPU or GCS, signs of sepsis or dehydration

### 2.3 Vital Signs and Monitoring

- Record SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature and blood glucose.
- Assess hydration and consciousness.

### 3. Management

#### 3.1 EMT-Level Management

- Reassure parents/carers.
- Oxygen therapy as needed.
- Encourage fluids if safe, oral paracetamol if able.
- Transport to paediatric facility if unwell or unclear source; if clear cause is present, treat as per relevant protocol.

#### 3.2 Paramedic-Level Management

- EMT-Level Management
- Administer antipyretics (paracetamol/ibuprofen)
- Monitor for febrile convulsions.
- Establish IV access only if required.

### 4. Key Points

- Fever is common in children and not always serious.
- Look for red flags: rash, seizures, lethargy, poor perfusion.
- Parental concern is an important indicator—take it seriously.







4



Trauma

**T1****MASS CASUALTY INCIDENTS**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for managing mass casualty incidents in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing mass casualty incidents in the prehospital setting.

#### 1.2 Definition

An MCI refers to any event that causes a significant number of casualties, overwhelming local emergency medical services (EMS) resources. The goal of the protocol is to ensure efficient triage, treatment, and transport of victims while maintaining effective coordination and resource management.

Mass Casualty Incidents (MCIs) can result from natural disasters, accidents, acts of terrorism, or other large-scale events, such as:

- **Natural Disasters:** Earthquakes, hurricanes, floods, wildfires.
- **Technological Disasters:** Fires, transportation accidents (e.g., bus/train crashes), hazardous materials spills.
- **Terrorism or Violence:** Active shooter events, bombings, biological, chemical, or radiological attacks.
- **Epidemics or Pandemics:** Large-scale outbreaks requiring mass triage and treatment.

During an MCI, multiple casualties require immediate medical attention, and EMS resources may be overwhelmed. It is essential to implement structured and systematic protocols to maximise the efficiency and effectiveness of care delivery.

## 2. Assessment

### 2.1 Initial Assessment

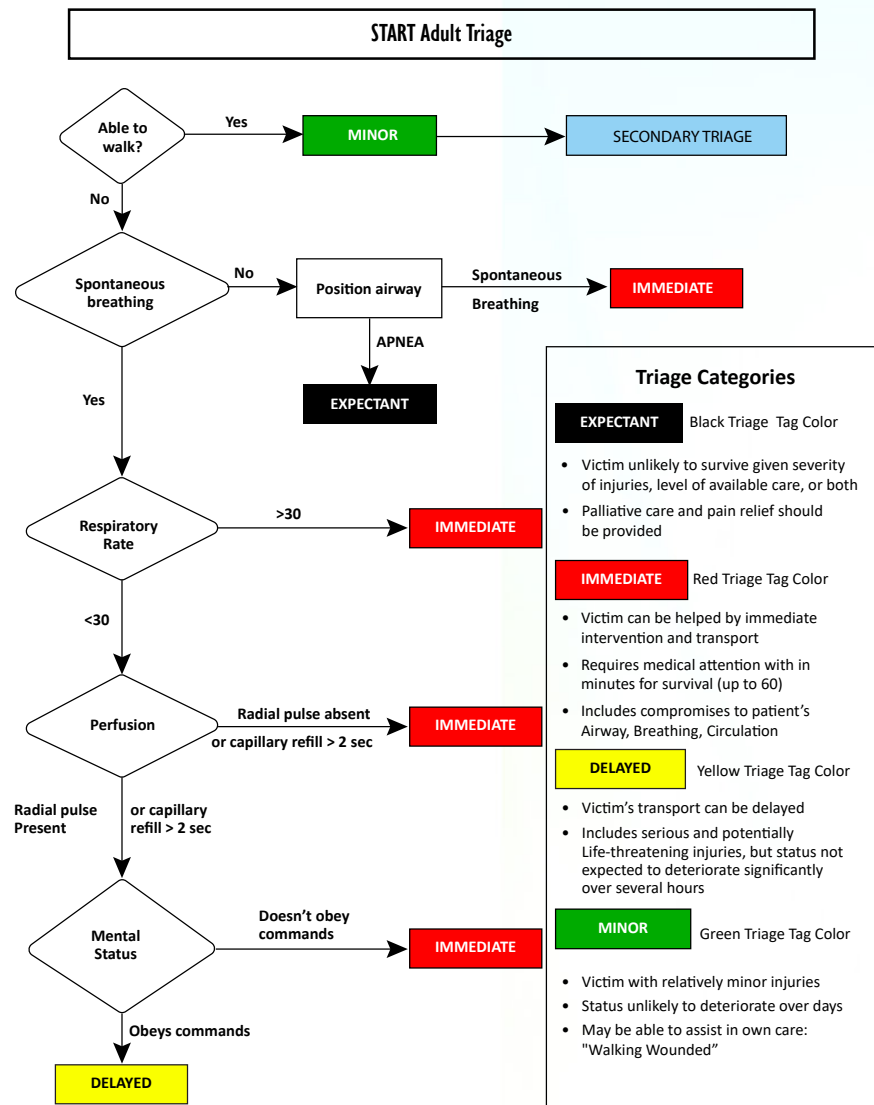
#### Scene Safety and Size-Up

- **Ensure Scene Safety:** Before approaching the incident, ensure the scene is secure. Assess for hazards such as fires, structural collapse, hazardous materials, or ongoing violence.
- **Establish Incident Command:** The Incident Commander (IC) should establish a unified command and coordinate the response efforts.
- **Assess the Scope:** Perform a rapid assessment of the incident site to estimate the number of casualties and the severity of injuries.

#### Triage Process

Triage is the process of sorting casualties based on the severity of their injuries, prioritising treatment for those with the greatest chance of survival. This should be done as quickly as possible to maximise the number of lives saved. Use a validated triage tool such as the START system.

- **START Triage System (Simple Triage and Rapid Treatment)** is commonly used:
  - **Green** (Minor): Patients with minor injuries who can walk and can self-treat or wait for care.
  - **Yellow** (Delayed): Patients with serious, but not life-threatening injuries. Treatment can be delayed for a short period.
  - **Red** (Immediate): Patients with life-threatening injuries who require immediate intervention to survive (e.g., massive haemorrhage, airway obstruction, or shock).
  - **Black** (Deceased/Expectant): Patients who are deceased or whose injuries are so severe that they are not expected to survive, even with immediate care.
- **Triage Tags:** Use triage tags or colour-coded identification to mark patients based on severity. Attach tags to a limb.
- During triage, the only initial interventions that should be completed are applying a CAT tourniquet and inserting an OPA. Once initial triage is complete, treat RED patients first.



## Resource Management and Coordination

### Incident Command System (ICS)

- Establish Unified Command:** An MCI requires coordinated efforts from multiple agencies (fire, law enforcement, hospitals). A unified command should be established to facilitate communication and coordination.
- Command Centre:** Set up a centralised location for coordinating efforts, allocating resources, and tracking patient status.
- Logistics:** Ensure that supplies such as medical equipment, medications, and personnel are appropriately distributed and replenished as needed.

## Medical Transport and Evacuation

- Transport Prioritisation:** Prioritise patients for transport based on their triage category and the available resources. Transport red-tagged patients first.
- Patient Distribution:** Coordinate with receiving hospitals to ensure appropriate patient distribution based on their severity and each facility's capabilities.
- Alternate Transport Methods:** If necessary, consider using helicopters, buses, or other mass transport options to evacuate patients quickly.

### 3. Management

#### 3.1 EMT-Level Management

Treat as per the relevant protocol for pathology.

#### 3.2 Paramedic-Level Management

Treat as per the relevant protocol for pathology.

### 4. Key Points

- Use a structured triage process that can be adopted locally.
- Ensure clear communication.
- Request assistance from the Police, the Kenyan Defence Force and the Fire Service where required.
- Treat specific pathology as per relevant protocol, most commonly Major Trauma.



**T2****MAJOR TRAUMA**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for managing major trauma in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing major trauma in the prehospital setting.



## 1.2 Definition

An injury or a combination of injuries that are life-threatening and could result in long-term disability or death.

**Patients with any of the following injuries should be treated as Major Trauma:**

- Severe head injury (e.g. depressed skull fracture, traumatic brain injury)
- Spinal injury with potential or actual cord involvement
- Chest injury (e.g. flail chest, open or tension pneumothorax, massive haemothorax)
- Abdominal injury with suspected internal bleeding
- Pelvic fracture with instability or suspected bleeding
- Two or more proximal long bone fractures
- Amputation of a limb
- Penetrating injuries to the head, neck, chest, abdomen, or groin
- Crush injury with prolonged entrapment
- Burns covering >15% TBSA in adults or >10% in children
- High-voltage electrical injury
- Near-drowning or hanging with injury
- High-energy transfer trauma (e.g. fall from height >2m, high-speed road traffic crash, pedestrian vs vehicle)
- Blast injuries

**Mechanism of Injury (MOI):**

While the mechanism alone does not define major trauma, certain mechanisms strongly indicate the potential for significant injury. These include:

- Ejection from the vehicle
- Fall from height >2 metres (or 3x body height in children)
- High-speed road traffic collision (>40 mph)
- Vehicle rollover or entrapment >30 minutes
- Death in the same vehicle
- Pedestrian or cyclist struck at speed
- Blast or explosion

**If any of the below physiology is present in a patient with any of the above mechanisms of injury, treat as Major Trauma:**

The presence of one or more of the following may indicate major trauma:

- Glasgow Coma Scale (GCS)  $\leq 13$
- Respiratory rate <10 or >29 breaths per minute
- Systolic blood pressure <90 mmHg (adults)
- Heart rate >120 bpm
- Evidence of shock or poor perfusion
- Uncontrolled external bleeding

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety and Initial Approach**
  - Haemorrhage Control
    - Apply direct pressure and wound packing for active bleeding.
    - Use haemostatic dressings and tourniquets when necessary.
    - Consider pelvic binders for suspected pelvic fractures.
- **Primary Survey (ABCDE Approach):**
  - **Airway:** Assess patency, airway management as required.
  - **Breathing:** Provide high-flow oxygen; assess for pneumothorax, haemothorax, or flail chest.
  - **Circulation:** Assess for signs of shock, BP and HR.
  - **Disability:** Evaluate neurological status using the Glasgow Coma Scale.
  - **Exposure:** Fully expose the patient to identify additional injuries while preventing hypothermia.

### 2.2 Focused Assessment

- **Focused History (SAMPLE)**
  - **S:** Symptoms such as pain or burns.
  - **A:** Note any allergies to medications or treatments.
  - **M:** Document current medications, including anticoagulants or eye drops.
  - **P:** Review past medical history.
  - **L:** Identify the last oral intake.
  - **E:** Describe the events leading to the injury.

### 2.3 Vital Signs and Monitoring

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

- Provide analgesia as required.**Extremity Trauma:**
  - Assess for fractures and vascular injury.
  - Immobilise fractures and monitor for compartment syndrome.
- Spinal Trauma:
  - Maintain spinal precautions.
  - Monitor for neurological deficits.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Advanced analgesia
- If hypotensive or suspected massive haemorrhage, initiate fluid resuscitation with 250ml boluses with a target BP of 90mmHg.
- IV Tranexamic acid 1g over 10 minutes
- Head and Neck Trauma:
  - Monitor for signs of increased intracranial pressure (ICP).
- Thoracic Trauma:
  - Perform needle decompression for tension pneumothorax.

### 4. Key Points

- If specific pathology is present, after following Major Trauma protocols, treat as per the relevant protocol during transport.
- Transport to a facility capable of trauma surgery if transport time is under 45 minutes and the patient is not peri-arrest.
- Paediatric Trauma: Adjust resuscitation based on age and weight.
- Pregnant Patients: Prioritise maternal stabilisation while monitoring foetal well-being.
- Elderly Trauma: Account for comorbidities and altered physiological response to injury. Consider Silver Trauma.

**T3****PENETRATING TRAUMA**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for managing penetrating trauma in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing penetrating trauma in the prehospital setting.

#### 1.2 Definition

Penetrating trauma is a type of injury where an object pierces the skin and enters the body, creating an open wound and potentially damaging underlying tissues, organs, or blood vessels.

Common causes include:

- Stab wounds (e.g. knives, broken glass)
- Gunshot wounds
- Impalement injuries
- Shrapnel or blast fragments

Characteristics:

- Visible entry wound (and sometimes exit wound)
- Localised damage along the object's path
- Risk of internal bleeding, infection, and organ injury

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety and Initial Approach**
  - Ensure personal safety and assess the scene for potential hazards (e.g., active threats).
  - Evaluate the mechanism of injury (e.g., location of wounds, type of weapon, or object involved).
- **Primary Survey (ABCDE)**
  - **Airway:**
    - Ensure airway patency.
    - Protect the cervical spine if spinal injury is suspected.
  - **Breathing:**
    - Assess respiratory rate, effort, and adequacy.
    - Look for signs of chest trauma (e.g., open pneumothorax, flail chest, or tension pneumothorax).
    - Administer high-flow oxygen if indicated.
- **Circulation:**
  - Identify and control major haemorrhage immediately.
  - Assess for signs of hypovolemic shock (e.g., weak pulses, tachycardia, hypotension).

## 2.2 Focused Assessment

- **Focused History (SAMPLE)**
  - **S:** Symptoms such as pain, difficulty breathing, or weakness.
  - **A:** Note any allergies to medications or treatments.
  - **M:** Document current medications, particularly anticoagulants.
  - **P:** Review past medical history, including prior injuries or surgeries.
  - **L:** Identify the patient's last oral intake.
  - **E:** Describe the events leading to the injury.
- **Focused Physical Examination**
  - Inspect for:
    - Entry and exit wounds.
    - Bleeding and signs of internal injury.
  - Palpate for tenderness, deformities, or crepitus.
  - Evaluate for:
    - Distended neck veins or tracheal deviation (suggesting tension pneumothorax).
    - Abdominal distention or rigidity (suggesting internal bleeding).
  - Monitor for signs of tamponade or shock (e.g., muffled heart sounds, hypotension).

## 2.3 Vital Signs and Monitoring

- Record SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.



### 3. Management

#### 3.1 EMT-Level Management

- **Haemorrhage Control**
  - Apply direct pressure to control bleeding.
  - Use haemostatic dressings or wound-packing techniques for junctional injuries.
  - For life-threatening extremity haemorrhage, apply a tourniquet proximal to the injury site. Note the time of application.
- **Chest Injuries**
  - For open chest wounds:
    - Apply a sterile occlusive dressing, securing it on three sides to create a three-way valve effect.
    - Monitor for signs of tension pneumothorax and release the dressing if necessary.
  - Treat suspected tension pneumothorax with needle decompression if within your scope of practice.
- **Abdominal Injuries**
  - Do not remove impaled objects; stabilise in place with bulky dressings.
  - Cover protruding organs (evisceration) with a moist, sterile dressing and avoid applying pressure.
- **Head and Neck Injuries**
  - Apply pressure carefully to avoid compromising the airway.
  - Stabilise impaled objects without removing them.
  - Monitor for signs of airway compromise or increased intracranial pressure.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Advanced analgesia
- Fluid resuscitation if the patient is hypotensive, aim for a systolic BP of 90 mmHg for chest injuries and 60- 70 mmHg for abdominal injuries.

#### 4. Key Points

- Transport all patients who are not peri-arrest to the closest hospital with surgical capability if within 45 minutes' drive time
- All central penetrating trauma, regardless of clinical presentation, should receive emergency transport to a surgical facility.

**T4****WOUND CLOSURE**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for wound closure in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing wound closure in the prehospital setting.

#### 1.2 Definition

Wound closure is essential to clinical practice; it plays a key role in the healing process and minimises complications such as infection, excessive scarring, or wound dehiscence. This protocol recommends appropriate methods, materials, and considerations for closing wounds, based on the latest evidence and expert consensus.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety and Initial Approach**
  - Ensure personal safety and assess for environmental hazards (e.g., chemical exposure, sharp objects).
  - Identify the mechanism of injury or condition (e.g., trauma, chemical exposure, spontaneous onset).
- **Primary Survey (ABCDE)**
  - **Airway:**
    - Ensure airway patency.
  - **Breathing:**
    - Assess respiratory status and administer oxygen as needed to maintain  $SpO_2 \geq 94\%$ .
  - **Circulation:**
    - Assess for associated injuries or systemic symptoms (e.g., shock, severe bleeding).

### 2.2 Focused Assessment

- **Focused History (SAMPLE)**
  - **S:** Symptoms such as pain, vision changes, or foreign body sensation.
  - **A:** Note any allergies to medications or treatments.
  - **M:** Document current medications, including anticoagulants or eye drops.
  - **P:** Review past medical history, including pre-existing eye conditions.
  - **L:** Identify the last oral intake.
  - **E:** Describe the events leading to the eye injury or condition.
- **Focused Physical Examination**
  - Inspect for redness, swelling, and tenderness at the site. If other relevant protocols apply, treat as per that protocol initially, with wound closure as a final management.

## 2.3 Vital Signs and Monitoring

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.

## 3. Management

### 3.1 EMT-Level Management

- EMTs may undertake the following under this protocol, provided the wound is clean, uncomplicated, and within the scope of training and resources available:
  - **Control bleeding** using direct pressure and appropriate dressings.
  - **Clean the wound** with clean water or sterile saline to remove visible debris.
  - **Assess wound depth, contamination, and location** to determine if closure is appropriate in the prehospital setting.
  - **Apply sterile adhesive strips** (e.g., Steri-Strips) to approximate wound edges for small, superficial lacerations.
  - **Cover with a sterile dressing** to protect the site.
  - **Reassess for signs of infection, bleeding, or compromised perfusion.**
  - **Pain management:** Administer paracetamol as per pain management protocol (if within local scope and protocols).
  - **Refer for further care** if:
    - The wound is deep, gaping, heavily contaminated, or involving high-risk areas (e.g., face, joints, hands, genitalia).
    - There is suspicion of underlying fractures, tendon/nerve damage, or foreign bodies.
    - There are signs of infection or systemic involvement.

### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Wound irrigation using large volumes of clean water or sterile saline under pressure for contaminated wounds.
- Closure assessment: Evaluate need for escalation to a higher level of care if wound is unsuitable for prehospital closure (e.g., deep lacerations, facial wounds, bites, puncture wounds).

#### 4. Key Points

- Wound closure can and should be completed in the prehospital setting by trained healthcare providers as a means of early management, hospital avoidance and maintenance of sterility.
- A sterile field is imperative to avoid infection
- Treat patients as per the relevant protocol initially, followed by wound closure guidance.

**T5****HEAD INJURY**

## Purpose

To provide EMTs and Paramedics with a standardised protocol for recognising and managing head injuries in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing head injuries in the prehospital setting.

#### 1.2 Definition

Any trauma with or without obvious injury to the head.



## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety and Initial Approach**
  - Ensure personal safety and scene security.
  - Identify the mechanism of injury (e.g., fall, motor vehicle collision, assault).
  - Look for potential hazards (e.g., unstable structures, debris).
- **Primary Survey (ABCDE)**
  - **Airway:** Assess patency. Be alert for blood, vomitus, or facial fractures that may compromise the airway.
    - Consider spinal precautions during airway management.
    - Use the jaw thrust manoeuvre if cervical spine injury is suspected.
  - **Breathing:** Evaluate respiratory rate, effort, and oxygen saturation.
    - Provide high-flow oxygen (10-15 L/min via non-rebreather mask) to maintain SpO<sub>2</sub> ≥ 94% or any critically unwell trauma patients.
  - **Circulation:** Assess for haemorrhage, pulse rate, blood pressure, and capillary refill.
    - Control scalp lacerations with direct pressure unless skull fracture is suspected.
  - **Neurological Assessment**
    - **Level of Consciousness (LOC):** Use the Glasgow Coma Scale (GCS) to assess responsiveness:
      - GCS 13-15: Mild brain injury.
      - GCS 9-12: Moderate brain injury.
      - GCS ≤8: Severe brain injury.
    - **Check Blood Glucose**
    - **Pupil Examination:**
      - Check for size, equality, and reactivity to light.



## 2.2 Focused Assessment

- **Focused History (SAMPLE)**
  - **S:** Symptoms such as headache, dizziness, confusion, or loss of consciousness.
  - **A:** Document allergies to medications or treatments.
  - **M:** List medications, especially anticoagulants or antiplatelets.
  - **P:** Explore past medical history, including prior head injuries or neurological conditions.
  - **L:** Identify the patient's last meal or substance intake.
  - **E:** Detail events leading to the injury (e.g., height of fall, speed of impact).
- **Secondary Survey**
  - Perform a head-to-toe examination, focusing on:
    - Scalp lacerations, deformities, or hematomas.
    - Battle's sign (bruising behind the ears) or raccoon eyes (periorbital bruising) indicate skull fracture.
    - Cerebrospinal fluid (CSF) leakage from the ears or the nose.
    - Motor and Sensory Function:
      - Assess for limb movement, strength, and sensation.
    - Signs of Increased Intracranial Pressure (ICP):
      - Headache, nausea/vomiting, altered LOC, unequal pupils, or Cushing's triad (hypertension, bradycardia, irregular respirations).
    - Signs of other injuries (e.g., cervical spine, chest, abdomen).

## 2.3 Vital Signs and Monitoring

- Record SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

- Arrest external haemorrhage with direct pressure.
- Provide oxygen if  $\text{SpO}_2 < 94\%$  or signs of hypoxia
- Maintain spinal alignment
- Consider providing pain relief if appropriate

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Provide IV/IM analgesia
- Consider advanced airway management if needed
- Haemorrhage control, including wound packing

### 4. Key Points

- Transport all patients with head injuries to the nearest emergency department with appropriate resources for trauma care.
- If the patient has a seizure, treat as per the seizure protocol.
- Priority transport for:
  - $\text{GCS} \leq 14$ .
  - Signs of increased ICP or herniation.
  - Persistent vomiting or worsening neurological status.
- Notify the receiving facility with:
  - Mechanism of injury.
  - Interventions performed and response to treatment.
  - Neurological findings and vital signs.

**T6****OPHTHALMIC INJURIES**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for managing ophthalmic injuries in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing ophthalmic injuries in the prehospital setting.

#### 1.2 Definition

An ophthalmic injury refers to any trauma or damage affecting the eye or its surrounding structures, including the eyelids, orbit (eye socket), conjunctiva, sclera, cornea, lens, retina, optic nerve, or extraocular muscles. These injuries can result from blunt force, penetrating trauma, chemical exposure, thermal burns, or radiation and may range in severity from minor irritation to vision-threatening conditions or permanent blindness.

In clinical settings, ophthalmic injuries are typically classified as:

- **Closed globe injuries** – where the outer wall of the eye remains intact (e.g., contusions, lamellar lacerations).
- **Open globe injuries** – involving a full-thickness wound of the eye wall (e.g., ruptures, penetrating injuries).

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety and Initial Approach**
  - Ensure personal safety and assess for environmental hazards (e.g., chemical exposure, sharp objects).
  - Identify the mechanism of injury or condition (e.g., trauma, chemical exposure, spontaneous onset).
- **Primary Survey (ABCDE)**
  - **Airway:**
    - Ensure airway patency; ophthalmic emergencies are rarely associated with airway compromise.
  - **Breathing:**
    - Assess respiratory status and administer oxygen as needed to maintain  $SpO_2 \geq 94\%$ .
  - **Circulation:**
    - Assess for associated injuries or systemic symptoms (e.g., shock, severe bleeding).

## 2.2 Focused Assessment

- **Focused History (SAMPLE)**
  - **S:** Symptoms such as pain, vision changes, or foreign body sensation.
  - **A:** Note any allergies to medications or treatments.
  - **M:** Document current medications, including anticoagulants or eye drops.
  - **P:** Review past medical history, including pre-existing eye conditions.
  - **L:** Identify the last oral intake.
  - **E:** Describe the events leading to the eye injury or condition.
- **Focused Physical Examination**
  - Inspect for:
    - Redness, swelling, or bruising around the eyes.
    - Blood in the anterior chamber (hyphaemia).
    - Foreign objects in or around the eye.
    - Signs of penetrating injury (e.g., distorted pupil, visible lacerations).
  - Assess vision:
    - Check visual acuity using a Snellen chart or by asking the patient to read text or count fingers.
    - Evaluate for double vision or loss of vision.
  - Palpate gently:
    - Avoid applying pressure to the globe.
    - Assess for bony tenderness around the orbit.

## 2.3 Vital Signs and Monitoring

- Record SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

- **General Management**
  - Avoid manipulation of the eye.
  - Do not apply pressure directly to the globe.
  - Cover both eyes to minimise movement if indicated (e.g., suspected globe rupture).
- **Specific Interventions**
  - **Chemical Exposure:**
    - Immediately irrigate the affected eye with sterile saline or water for at least 15-20 minutes.
    - Continue irrigation during transport.
    - Identify the chemical, if possible, and bring the container or Safety Data Sheet (SDS) to the hospital.
  - **Foreign Body (Superficial):**
    - Do not attempt to remove embedded foreign bodies.
    - For loose debris, rinse with sterile saline.
  - **Penetrating Eye Injuries:**
    - Do not remove impaled objects.
    - Stabilise the object with a bulky dressing.
    - Cover the unaffected eye to reduce movement.
  - **Blunt Trauma:**
    - Protect the eye with a rigid shield (e.g., cup or specialised eye shield).
    - Monitor for signs of orbital fracture or increased intraocular pressure (e.g., severe pain, proptosis).
  - **Acute Vision Loss:**
    - Assess for stroke symptoms if vision loss is sudden and unilateral.
    - Manage other associated symptoms per local protocols.

- **Infection Prevention**
  - Use sterile dressings for open wounds.
  - Avoid introducing contaminants into the eye.

### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Advanced analgesia

### 4. Key Points

- Transport to a facility capable of ophthalmic management if complete vision loss is present and the transport time is under 45 minutes.
- All other patients should be transported to the closest emergency department.

## T7

## SPINAL INJURY

## Purpose

To provide EMTs and Paramedics with a standardised protocol for recognising and managing spinal injuries in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing spinal injuries in the prehospital setting.

#### 1.2 Definition

**Spinal injury** refers to actual or suspected trauma to the spinal column and/or spinal cord resulting from a mechanism of injury with the potential to cause structural damage. This may involve vertebral fractures, dislocations, ligamentous injury, or direct insult to the spinal cord.

In the pre-hospital setting, a spinal injury should be **suspected** in any patient who has sustained a significant mechanism of injury with signs or symptoms suggestive of spinal cord involvement, such as:

- Pain or tenderness along the spine
- Neurological deficits (e.g. numbness, weakness, paralysis)
- Altered sensation or proprioception.
- Loss of bladder or bowel control
- Abnormal posturing or spinal deformity



**High-risk mechanisms include:**

- Falls from height.
- High-speed road traffic collisions
- Diving injuries
- Blunt trauma to the head, neck, or back
- Ejection from a vehicle
- Pedestrian or cyclist struck by a vehicle
- Gunshot wound to chest or abdomen



## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety and Initial Approach**
  - Ensure personal safety and assess the environment for potential hazards.
  - Identify the mechanism of injury (MOI), such as falls, motor vehicle collisions, or penetrating trauma.
  - Assume spinal injury in high-risk mechanisms until ruled out.
- **Primary Survey (ABCDE)**
  - **Airway:** Maintain airway patency using a jaw thrust manoeuvre if spinal injury is suspected.
    - Avoid hyperextension or manipulation of the neck.
    - Consider advanced airway management if required.
  - **Breathing:** Assess for adequate ventilation and oxygenation.
    - Administer high-flow oxygen to maintain SpO<sub>2</sub> ≥94%.
    - Be alert for signs of respiratory compromise due to high cervical spinal injury (e.g., C2-C5 involvement).
  - **Circulation:** Evaluate heart rate, blood pressure, and signs of shock.
    - Control external haemorrhage without excessive movement of the spine.
  - **Neurological Assessment**
    - **Level of Consciousness (LOC):** Use the Glasgow Coma Scale (GCS) to assess responsiveness:
      - GCS 13-15: Mild brain injury.
      - GCS 9-12: Moderate brain injury.
      - GCS ≤8: Severe brain injury.
    - **Check Blood Glucose**
    - **Pupil Examination:**
      - Check for size, equality, and reactivity to light.

## 2.2 Focused Assessment

- **Focused History (SAMPLE)**
  - **S:** Symptoms such as pain, numbness, or paralysis.
  - **A:** Note allergies to medications or treatments.
  - **M:** Document current medications, especially anticoagulants.
  - **P:** Review medical history for conditions affecting the spine (e.g., prior surgery, osteoporosis).
  - **L:** Identify the patient's last meal or substance intake.
  - **E:** Detail the events leading to the injury, including the MOI.
- **Neurological Assessment**
  - Assess level of consciousness using the Glasgow Coma Scale (GCS).
  - Perform a detailed neurological exam:
    - **Motor Function:** Assess for voluntary movement in all extremities.
    - **Sensory Function:** Check for sensation to light touch and pinprick.
    - **Reflexes:** Observe for abnormal reflexes, such as priapism or Babinski sign.
  - Document any neurological deficits.
- **Spinal Assessment**
  - Palpate the entire spine for:
    - Pain or tenderness.
    - Deformity or step-offs.
    - Muscle spasm.
  - Observe for signs of spinal injury:
    - Incontinence.
    - Flaccid paralysis.
    - Neurogenic shock (hypotension with bradycardia).

## 2.3 Vital Signs and Monitoring

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.

## 3. Management

### 3.1 EMT-Level Management

- Immobilise all patients with any subjective or objective reports of neurological deficit, midline spinal pain, altered GCS or concerning mechanism of injury. Patients should not routinely be left on spine boards for more than 45 minutes if possible.
- Provide oxygen and pain relief as indicated

### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Advanced analgesia if required.

## 4. Key Points

- A high force is required to cause spinal injury; therefore, in patients where this is suspected, there are often other serious injuries that are concurrent with the spinal injury.
- Transport to hospital capable of trauma surgery if within 45 minutes ' drive time, otherwise transport to the closest emergency department.

**T8****CHEST INJURY**

## Purpose

To provide EMTs and Paramedics with a standardised protocol for recognising and managing chest injury in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing chest injuries in the prehospital setting.

#### 1.2 Definition

A **chest injury** refers to any form of trauma that affects the chest area, including the skin, ribs, sternum, muscles, and the organs within the thoracic cavity such as the lungs, heart, and major blood vessels. Chest injuries can range from minor to life-threatening and are commonly classified into two main types:

- **Blunt chest injury** – caused by impact from a force without penetration (e.g. falls, road traffic collisions, assaults).
- **Penetrating chest injury** – where an object pierces the chest wall (e.g. stab wounds, gunshot wounds).

Examples of chest injuries:

- Rib fractures
- Flail chest
- Pneumothorax (air in the pleural space)
- Haemothorax (blood in the pleural space)
- Pulmonary contusion
- Cardiac tamponade
- Sternal fracture

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety and Initial Approach**
  - Ensure personal safety and assess the scene for hazards.
  - Determine the mechanism of injury (e.g., motor vehicle collision, fall, assault, penetrating trauma).
- **Primary Survey (ABCDE)**
  - **Airway:**
    - Assess airway patency and address obstruction (e.g., blood, vomitus, or foreign bodies).
    - Use a jaw thrust manoeuvre if spinal injury is suspected.
    - Consider advanced airway management if necessary.
  - **Breathing:**
    - Evaluate respiratory rate, depth, effort, and symmetry of chest rise.
    - Auscultate for breath sounds bilaterally.
    - Identify signs of respiratory distress:
      - Cyanosis, tachypnoea, use of accessory muscles, or tracheal deviation.
    - Administer high-flow oxygen (10-15 L/min via non-rebreather mask) to maintain SpO<sub>2</sub> ≥94%.
  - **Circulation:**
    - Assess for signs of shock (e.g., hypotension, tachycardia, pale or cool skin).
    - Evaluate for external bleeding and control haemorrhage.
    - Palpate for chest wall tenderness, deformities, or crepitus.

## 2.2 Focused Assessment

- **Focused History (SAMPLE)**
  - **S:** Symptoms such as chest pain, difficulty breathing, or referred pain (e.g., to the shoulder or back).
  - **A:** Note allergies to medications or treatments.
  - **M:** Document current medications, particularly anticoagulants.
  - **P:** Review past medical history, including prior chest injuries or lung conditions.
  - **L:** Identify the patient's last meal or substance intake.
  - **E:** Detail the events leading to the injury, including the MOI.
- **Focused Physical Examination**
  - Inspect the chest for:
    - Open wounds (e.g., sucking chest wounds).
    - Bruising, deformity, or paradoxical chest wall motion (flail chest).
  - Palpate the chest for:
    - Tenderness, deformities, or subcutaneous emphysema.
  - Auscultate for:
    - Diminished or absent breath sounds.
    - Additional sounds such as crackles or wheezes.

## 2.3 Vital Signs and Monitoring

- Record SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

- **Airway and Breathing Management**
  - Secure the airway as needed, ensuring minimal movement of the cervical spine if spinal injury is suspected.
  - Provide high-flow oxygen to improve oxygenation.
  - Address specific conditions:
    - **Tension Pneumothorax:**
      - Perform needle decompression at the second intercostal space, midclavicular line, or fifth intercostal space, anterior to the midaxillary line.
      - Monitor for improved ventilation and hemodynamic stability.
    - **Open Pneumothorax (Sucking Chest Wound):**
      - Apply a sterile occlusive dressing secured on three sides to act as a one-way valve.
    - **Flail Chest:**
      - Provide high-flow oxygen and consider positive pressure ventilation if indicated. Support the chest wall with a large dressing.
- Arrest external haemorrhage with haemostatic gauze.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Fluid resuscitation if hypotensive
- Advanced analgesia
- IV/IM Tranexamic Acid 1g if major, ongoing, external or internal haemorrhage is suspected.



#### 4. Key Points

- Transport all patients with significant chest injuries to a hospital with surgical capabilities.
- Prioritise rapid transport for:
  - Tension pneumothorax or suspected haemothorax.
  - Flail chest with respiratory compromise.
  - Cardiac tamponade.
  - Persistent hypotension or shock.



**T9****PELVIC INJURIES**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for managing pelvic injuries in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing pelvic injuries in the prehospital setting.

#### 1.2 Definition

Pelvic injury refers to any trauma involving the bony pelvis or the organs and structures contained within or surrounding the pelvic region, such as blood vessels, bladder, bowel, and reproductive organs. Of most concern are pelvic fractures, which are associated with significant mechanisms of injury and are life-threatening.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety and Initial Approach**
  - Ensure personal safety and assess for environmental hazards.
  - Determine the mechanism of injury and assess for concurrent injuries.
- **Primary Survey (ABCDE)**
  - **Airway:**
    - Ensure airway patency.
    - Protect the cervical spine if indicated by the mechanism of injury.
  - **Breathing:**
    - Assess respiratory rate, effort, and oxygenation.
    - Administer high-flow oxygen to maintain SpO<sub>2</sub> ≥94%.
  - **Circulation:**
    - Identify and control life-threatening haemorrhage.
    - Assess for signs of hypovolemic shock (e.g., tachycardia, hypotension, altered mental status).

## 2.2 Focused Assessment

- **Focused History (SAMPLE)**
  - **S:** Symptoms such as pelvic pain, difficulty moving, or numbness in the lower extremities.
  - **A:** Note any allergies to medications or treatments.
  - **M:** Document current medications, particularly anticoagulants.
  - **P:** Review past medical history, including previous pelvic or orthopaedic injuries.
  - **L:** Identify the last oral intake.
  - **E:** Describe the events leading to the injury.
- **Focused Physical Examination**
  - Inspect for:
    - Obvious deformities, bruising, or swelling in the pelvic region.
    - Bleeding from the urethra, rectum, or vagina (indicative of severe injury).
  - **Palpate gently:**
    - Avoid excessive movement to prevent exacerbating internal bleeding or injury. DO NOT FLEX THE PELVIS as this can increase vascular damage.
    - Assess for instability (e.g., crepitus or abnormal mobility) only if necessary.
  - **Evaluate for:**
    - Shortening or rotation of the lower extremities.
    - Signs of neurovascular compromise in the legs (e.g., absent pulses, altered sensation).

## 2.3 Vital Signs and Monitoring

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

- **Pelvic Stabilisation**
  - **Pelvic Binder:**
    - Apply a commercial pelvic binder or improvised sling at the level of the greater trochanters (not the iliac crests).
    - Do not delay transport to apply the binder.
  - If a binder is unavailable:
    - Use a sheet or similar material to stabilise the pelvis, ensuring adequate tension to reduce movement.
- **Pain Management**
  - Administer analgesics as needed.
- **Shock Management**
  - Establish IV/IO access and initiate fluid resuscitation for signs of shock:
    - Adults: 250- 500 ml of isotonic crystalloids (e.g., normal saline or lactated Ringer's).
    - Paediatrics: 20 ml/kg IV bolus.
  - Avoid excessive fluids if haemorrhage is suspected; maintain systolic BP at 80-90 mmHg (permissive hypotension) unless traumatic brain injury is suspected.
- **Immobilisation and Transport**
  - Avoid unnecessary movement of the patient.
  - Immobilise the patient on a backboard or vacuum mattress as indicated.
  - Secure the pelvic binder in place prior to movement or transport.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Advanced analgesia

### 4. Key Points

- Transport all patients who are not peri-arrest to the closest hospital with surgical capability if within 45 minutes' drive time.

## T10

## LIMB INJURIES

## Purpose

To provide EMTs and Paramedics with a standardised protocol for recognising and managing limb injuries in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing limb injuries in the prehospital setting.

#### 1.2 Definition

A **limb injury** refers to any damage or trauma affecting the upper limbs (shoulder, arm, elbow, forearm, wrist, hand) or lower limbs (hip, thigh, knee, leg, ankle, foot). These injuries can vary widely in severity from minor to life-threatening and may involve:

- **Bones** – fractures or dislocations
- **Muscles and tendons** – strains or tears
- **Ligaments** – sprains or ruptures
- **Blood vessels** – bleeding or compromised circulation
- **Nerves** – numbness, weakness, or loss of function
- **Skin and soft tissue** – abrasions, lacerations, or crush injuries

Common causes include:

- Falls
- Sports injuries
- Road traffic collisions
- Direct blows or crush injuries
- Penetrating trauma (e.g. stab wounds)

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety and Initial Approach**
  - Ensure personal safety and assess the environment for hazards.
  - Identify the mechanism of injury (e.g., fall, motor vehicle collision, crush injury, or penetrating trauma).
- **Primary Survey (ABCDE)**
  - **Airway:**
    - Ensure airway patency and address obstruction if present.
    - Use spinal precautions if a high-risk mechanism is suspected.
  - **Breathing:**
    - Ensure adequate ventilation and oxygenation. Administer high-flow oxygen as needed.
  - **Circulation:**
    - Control significant external bleeding with direct pressure, haemostatic dressings, or tourniquets as appropriate.
    - Assess for signs of hypovolemic shock (e.g., tachycardia, hypotension, or pallor).

## 2.2 Focused Assessment

- **Focused History (SAMPLE)**
  - **S:** Symptoms such as pain, numbness, or inability to move the affected limb.
  - **A:** Note any allergies to medications or treatments.
  - **M:** Document current medications, especially anticoagulants.
  - **P:** Review past medical history, including prior injuries or surgeries to the limb.
  - **L:** Identify the patient's last meal or substance intake.
  - **E:** Detail the events leading to the injury.
- **Focused Physical Examination**
  - Inspect the affected limb for:
    - Deformities, swelling, bruising, or open wounds.
    - Shortening, angulation, or rotation of the limb.
  - Palpate for tenderness, crepitus, or abnormal movement.
  - Assess neurovascular status:
    - **Circulation:** Check distal pulses, capillary refill, and skin colour.
    - **Sensation:** Test for light touch and pain sensation distal to the injury.
    - **Motor Function:** Assess the patient's ability to move fingers or toes.
  - Monitor for signs of compartment syndrome:
    - Pain is out of proportion to the injury, pallor, pulselessness, paraesthesia, or paralysis.

## 2.3 Vital Signs and Monitoring

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.



### 3. Management

#### 3.1 EMT-Level Management

- Arrest external haemorrhage with haemostatic gauze
- Splint suspected fractures in the position of comfort, only straighten if neurovascularly compromised.
- Provide analgesia.
- Wound management as per wound closure protocols.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Advanced analgesia
- IV/IM Tranexamic Acid 1gm if major, ongoing, external or internal haemorrhage is suspected

### 4. Key Points

- Transport all patients with significant limb injuries to a hospital with surgical capabilities.
- In the event of an amputation, follow amputation protocols.

## T11

## AMPUTATION

## Purpose

To provide EMTs and Paramedics with a standardised protocol for managing amputations in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing limb injuries in the prehospital setting.

#### 1.2 Definition

**Amputation** is the **traumatic removal of all or part of a limb or extremity**, such as an arm, leg, hand, foot, finger, or toe. It typically results from:

- Accidents or severe injuries
- Combat-related trauma

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety and Initial Approach**
  - Ensure personal safety and assess the environment for hazards.
  - Identify the mechanism of injury and assess the potential for additional injuries.
- **Primary Survey (ABCDE)**
  - **Airway:**
    - Ensure airway patency and address obstructions as needed.
    - Use spinal precautions if indicated by the mechanism of injury.
  - **Breathing:**
    - Assess respiratory rate, effort, and oxygenation.
    - Administer high-flow oxygen to maintain  $SpO_2 \geq 94\%$ .
  - **Circulation:**
    - Assess for life-threatening haemorrhage and control bleeding immediately.
    - Evaluate for signs of hypovolemic shock (e.g., tachycardia, hypotension, altered mental status).

## 2.2 Focused Assessment

- **Focused History (SAMPLE)**
  - **S:** Symptoms such as pain, numbness, or difficulty moving the affected area.
  - **A:** Note any allergies to medications or treatments.
  - **M:** Document current medications, particularly anticoagulants.
  - **P:** Review past medical history, including prior injuries or surgeries.
  - **L:** Determine the last oral intake.
  - **E:** Describe the events leading to the amputation, including the mechanism of injury.
- **Focused Physical Examination**
  - Inspect the injury site:
    - Confirm if the amputation is partial or complete.
    - Note the location and extent of tissue loss.
- Assess for:
  - Active bleeding.
  - Exposed bone, nerves, or vascular structures.
- Evaluate distal neurovascular status (for partial amputations):
  - Check pulses, capillary refill, sensation, and motor function.

## 2.3 Vital Signs and Monitoring

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

- **Haemorrhage Control**
  - Apply direct pressure to control bleeding.
  - Use a pressure dressing or haemostatic agents if bleeding persists.
  - If direct pressure is insufficient or the bleeding is life-threatening:
    - Apply a tourniquet proximal to the amputation site.
    - Note the time of application and ensure it is visible.
  - Reassess frequently to ensure bleeding is controlled.
  - 1gm Tranexamic Acid IV over 10 minutes/IM
- **Pain Management**
  - Administer analgesics as needed.
- **Management of the Amputated Part**
  - Locate the amputated part and handle it gently:
    - Remove gross contaminants by rinsing with sterile saline.
    - Wrap the part in sterile gauze moistened with saline.
    - Place the wrapped part in a waterproof bag.
    - Keep the bag cool by placing it in a container with ice or cold packs (avoid direct contact with ice to prevent frostbite).
  - Label the bag with the patient's name, the time of injury, and the type of amputation.
- **Stabilisation and Transport**
  - Splint the affected extremity to minimise movement and prevent further injury.
  - Cover the wound with a sterile dressing and bandage.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Advanced analgesia

#### 4. Key Points

- Transport all patients with amputations to a facility capable of trauma surgery if within 45 45-minute transport time
- Always keep the amputated part with the patient

**T12****BURNS**

## Purpose

This protocol provides EMTs and Paramedics with a standardised approach for managing burns in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing burns in the prehospital setting.

## 1.2 Definition

A burn is an injury to the skin and/or underlying tissues caused by heat, chemicals, electricity, sunlight, or radiation. Burns are classified based on their depth.

Burn Depth Classification:

- **Superficial (First-degree):**
  - Affects only the epidermis
  - Red, painful, dry (e.g., mild sunburn)
  - No blisters, heals without scarring
- **Partial Thickness (Second-degree):**
  - Involves the epidermis and part of the dermis
  - Blistering, redness, swelling, and moist appearance
  - Very painful; may scar
- **Full Thickness (Third-degree):**
  - Destroys epidermis and dermis, may extend to subcutaneous tissue
  - Skin may appear white, leathery, or charred
  - Painless (nerve endings destroyed), requires surgical intervention
- **Fourth degree (sometimes used):**
  - Extends into muscle, bone, or tendons
  - Typically results from high-voltage electrical burns or severe thermal injuries



## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure the environment is safe for rescuers and patients.
- **Triage:** Prioritise based on burn severity and associated trauma.
- **Primary Survey (ABCDE Approach):**
  - **Airway:** Assess for inhalation injury, facial burns, or airway compromise.
  - **Breathing:** Provide oxygen therapy if respiratory distress is present.
  - **Circulation:** Evaluate perfusion, assess for shock, and control any bleeding.
  - **Disability:** Check neurological status using the Glasgow Coma Scale.
  - **Exposure:** Remove constrictive clothing and assess burn depth and total body surface area (TBSA) using the rule of 9s or the hand method.

### 2.2 Focused Assessment

- Focused History (SAMPLE)
  - **S:** Symptoms such as pain.
  - **A:** Note any allergies to medications or treatments.
  - **M:** Document current medications, including anticoagulants or eye drops.
  - **P:** Review past medical history, including pre-existing eye conditions.
  - **L:** Identify the last oral intake.
  - **E:** Describe the events leading to injury.

### 2.3 Vital Signs and Monitoring

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.

## 3. Management

### 3.1 EMT-Level Management

- **Cooling:** Apply cool (not ice-cold) running water for 20 minutes within the first three hours of injury. THIS SHOULD BE PRIORITISED OVER TRANSPORT
- **Pain Management:**
  - Mild burns: Oral analgesics (acetaminophen, NSAIDs).
- **Oxygen:** Administer oxygen via a non-rebreather mask to all patients, as SPO2 may not be accurate due to CO poisoning.
- **Infection Control:**
  - Cover burns with non-adherent dressings.
  - Avoid rupturing blisters unless infected.
- **Transport Considerations:**
  - Expedite transport to an appropriate facility.

### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Advanced analgesia
- Fluid Resuscitation:
  - Use Parkland formula:  $4 \text{ mL} \times \text{body weight (kg)} \times \% \text{TBSA burned} = \text{total fluid}$  (half given in the first 8 hours, remainder over the next 16 hours).
  - Monitor urine output (target: 0.5-1 mL/kg/hr in adults, 1-2 mL/kg/hr in children).

#### 4. Key Points

- **Inhalation Injury:**
  - Look for soot in the airway, facial burns, and hoarseness.
  - Provide high-flow oxygen and prepare for rapid transport.
- **Electrical Burns:**
  - Assess for cardiac arrhythmias.
- **Chemical Burns:**
  - Irrigate with copious amounts of water (except for alkali metals, which require specific decontamination).
  - Paediatric and Geriatric Patients:
- **Adjust assessment and interventions accordingly.**
- **Transfer Criteria to Advanced Burns Care over local hospital, if transport time less than 45mins**
  - Burns >10% TBSA in patients <10 or >50 years old.
  - Burns >20% TBSA in other age groups.
  - Full-thickness burns >5% TBSA.
  - Burns involving the face, hands, feet, genitals, perineum, or major joints.
  - Inhalation injury, electrical or chemical burns, or significant comorbidities.

## T13

## ELECTRIC SHOCK

## Purpose

To provide EMTs and Paramedics with a standardised protocol for managing electric shock in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing electric shock in the prehospital setting.

#### 1.2 Definition

An electric shock is a physiological reaction or injury caused by the passage of an electric current through the body. This can occur when a person comes into direct contact with an electrical energy source, allowing current to flow through tissues, potentially disrupting neurological, muscular, and cardiac functions.

- **Low voltage (< 1,000v):** More common in household and workplace incidents.
- **High voltage (> 1,000v):** Associated with industrial accidents, power lines, and lightning strikes.
- **Lightning Injuries:** Unique mechanisms with potential for blunt trauma and multisystem effects.

## 2. Assessment

### 2.1 Initial Assessment

- Scene Safety and Initial Approach
  - Ensure personal safety and assess for environmental hazards (e.g., chemical exposure, sharp objects).
  - Identify the mechanism of injury or condition (e.g., trauma, chemical exposure, spontaneous onset).
- Primary Survey (ABCDE)
  - **Airway:** Ensure airway patency; ophthalmic emergencies are rarely associated with airway compromise.
  - **Breathing:** Assess respiratory status and administer oxygen as needed to maintain SpO<sub>2</sub> ≥ 94%.
  - **Circulation:** Assess for associated injuries or systemic symptoms (e.g., shock, severe bleeding).

### 2.2 Focused Assessment

- Focused History (SAMPLE)
  - **S:** Symptoms such as pain or burns.
  - **A:** Note any allergies to medications or treatments.
  - **M:** Document current medications, including anticoagulants or eye drops.
  - **P:** Review past medical history.
  - **L:** Identify the last oral intake.
  - **E:** Describe the events leading to the injury.

### 2.3 Vital Signs and Monitoring

- Record SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

- Cardiac Monitoring:
  - Perform ECG to detect arrhythmias (e.g., ventricular fibrillation, asystole, QT prolongation).
  - Continuous monitoring for delayed cardiac effects, especially in high-voltage injuries.
- Pain Management:
  - Mild: NSAIDs or acetaminophen.
- Burn and Wound Care:
  - Cover burns with sterile dressings.
  - Monitor for compartment syndrome in extremities.
  - Avoid excessive cooling, which may induce hypothermia.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Advanced analgesia
- Fluid Resuscitation:
  - IV access and aggressive fluid management for burns and rhabdomyolysis prevention.
  - Monitor urine output (target: 1-2 ml/kg/hr) to prevent acute kidney injury.

### 4. Key Points

- Consider other presenting or sequela pathology and treat as per relevant protocols.

**T14****DROWNING**

## Purpose

To provide EMTs and Paramedics with a standardised protocol for managing drowning in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing drowning in the prehospital setting.

#### 1.2 Definition

**Drowning** is the process of experiencing respiratory impairment due to submersion or immersion in a liquid.

#### Submersion vs Immersion

- **Submersion:** The airway is entirely under the surface of the liquid.
- **Immersion:** The airway is at the surface or intermittently under the liquid (e.g. face splashing or partial airway obstruction).

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure a safe rescue environment to prevent additional casualties.
- **Primary Survey (ABCDE Approach):**
  - **Airway:** Clear obstructions, consider early intubation if necessary.
  - **Breathing:** Provide oxygen; assess for aspiration and pulmonary oedema.
  - **Circulation:** Check for pulses; initiate CPR if needed.
  - **Disability:** Assess neurological status with Glasgow Coma Scale (GCS).
  - **Exposure:** Prevent hypothermia with appropriate warming techniques.
- If cardiac arrest, treat as per the Cardiac Arrest Protocol.

### 2.2 Focused Assessment

- **Focused History (SAMPLE)**
  - **S:** Symptoms such as pain.
  - **A:** Note any allergies to medications or treatments.
  - **M:** Document current medications.
  - **P:** Review past medical history.
  - **L:** Identify the last oral intake.
  - **E:** Describe the events leading to the drowning.

### 2.3 Vital Signs and Monitoring

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.



### 3. Management

#### 3.1 EMT-Level Management

- Provide positive pressure ventilation if required
- Passive or active warming, if possible, with a heater or warmed fluids if hypothermic
- Provide high-flow oxygen if hypoxic

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Advanced airway if unconscious

### 4. Key Points

- **Paediatric Patients:** Special attention to hypothermia and airway management.
- **Cold-Water Drowning:** Prolonged resuscitation efforts may be beneficial.
- **Near-Drowning:** Observe for delayed complications like pulmonary oedema.

5

Environmental



**E1****HYPOTHERMIA**

## Purpose

To provide EMTs and Paramedics with a standardised protocol for managing hypothermia in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing hypothermia in the prehospital setting.

#### 1.2 Definition

Hypothermia is a core body temperature of  $< 35^{\circ}\text{C}$  or  $95^{\circ}\text{F}$ .

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure responder safety and prevent further exposure.
- **Primary Survey (ABCDE Approach):**
  - **Airway:** Assess for patency; be cautious of decreased protective reflexes.
  - **Breathing:** Assess respiratory effort; provide oxygen as needed.
  - **Circulation:** Check for pulses carefully as bradycardia may be present.
  - **Disability:** Assess neurological status using Glasgow Coma Scale (GCS).
  - **Exposure:** Remove wet clothing and cover the patient with insulating materials.

### 2.2 Focused Assessment

- Classification of Hypothermia
  - **Mild (32-35°C / 90-95°F):** Shivering, tachycardia, normal mental status.
  - **Moderate (28-32°C / 82-90°F):** Reduced shivering, confusion, bradycardia.
  - **Severe (<28°C / <82°F):** Unconsciousness, arrhythmias, risk of cardiac arrest.

### 2.3 Vital Signs and Monitoring

- Record SPO2, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

- **Prevent Further Heat Loss:**
  - Remove the patient from the cold environment.
  - Replace wet clothing with dry, insulated layers.
  - Use warm blankets and vapor barriers to retain body heat.
- **Rewarming Strategies:**
  - **Mild Hypothermia:** Apply passive external warming (blankets, dry clothing, warm environment).
  - **Moderate Hypothermia:** Use active external warming (warm packs to the chest, armpits, and groin).
  - **Severe Hypothermia:** Focus on core rewarming; avoid aggressive movement to prevent arrhythmias.
- **Cardiac Monitoring:**
  - Handle the patient gently to prevent triggering ventricular fibrillation.
  - Check pulse carefully, as bradycardia may be mistaken for pulselessness.
  - Initiate CPR only if no signs of life are detected after careful assessment.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Advanced airway if unconscious and required
- Fluid Resuscitation:
  - Provide warm IV fluids if available and indicated.
  - Avoid excessive fluid administration to prevent complications.

#### 4. Key Points

- **Cold Water Submersion:** Consider prolonged resuscitation in cases of cold-water drowning.
- **Elderly and Paediatric Patients:** Higher risk due to impaired thermoregulation.
- **Trauma Patients:** Hypothermia worsens coagulopathy; prioritise warming.
- **Transport Considerations:**
  - Keep the patient warm during transport.
  - Minimise rough handling to prevent cardiac disturbances.
- If hypothermia is a sequela condition, treat as per the relevant protocol also

**E2****HYPERTHERMIA**

## Purpose

To provide EMTs and Paramedics with a standardised protocol for managing hyperthermia in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing hyperthermia in the prehospital setting.

#### 1.2 Definition

Hyperthermia is defined as an abnormally elevated body temperature due to failed thermoregulation, where the body absorbs or produces more heat than it can dissipate.

Unlike fever, which is a regulated rise in body temperature due to infection, hyperthermia is unregulated and can be life-threatening.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Remove the patient from the hot environment.
- **Primary Survey (ABCDE Approach):**
  - **Airway:** Assess patency.
  - **Breathing:** Provide supplemental oxygen if needed.
  - **Circulation:** Assess for dehydration, hypotension, or shock.
  - **Disability:** Evaluate neurological status using the Glasgow Coma Scale (GCS).
  - **Exposure:** Remove excess clothing to facilitate cooling.

### 2.2 Focused Assessment

#### Classification of Hyperthermia:

|                                     |           |                                                          |
|-------------------------------------|-----------|----------------------------------------------------------|
| Mild hyperthermia (Heat exhaustion) | 38–39°C   | Flushed skin, sweating, tachycardia, and heat discomfort |
| Moderate hyperthermia (Heat Stroke) | 39–40.9°C | Fatigue, nausea, dizziness, and confusion                |
| Severe hyperthermia (Heat Stroke)   | ≥41°C     | Altered mental status, seizures, hypotension, coma       |

### 2.3 Vital Signs and Monitoring

- Record SP02, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.



### 3. Management

#### 3.1 EMT-Level Management

- **Remove from Heat Exposure:**
  - Move the patient to a shaded, cool, or air-conditioned environment.
  - Loosen or remove restrictive clothing.
- **Initiate Cooling Measures:**
  - **Mild to Moderate Hyperthermia:** Oral rehydration with cool fluids if the patient is alert and not vomiting.
  - **Severe Hyperthermia:**
    - Rapid evaporative cooling (spray cool water and fan air across the patient).
    - Apply ice packs to the neck, axillae, and groin.
    - Immerse in cool or ice water if feasible and safe.
- **Hydration and Circulation Support:**
  - Establish IV access for fluid resuscitation with cold saline if available.
  - Monitor for signs of shock and organ dysfunction.
- **Neurological and Cardiac Monitoring:**
  - Monitor for confusion, seizures, or decreased responsiveness.
  - Be prepared for potential cardiac arrhythmias.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Advanced airway if unconscious and required
- Anticonvulsant if seizure persists, treat as per seizure protocol.

#### 4. Key Points

- **Elderly and Paediatric Patients:** Higher risk due to impaired thermoregulation.
- **Exertional Heat Stroke:** More common in athletes and labourers; requires aggressive cooling.
- **Medication-Related Hyperthermia:** Consider drug-induced causes (e.g., stimulants, antipsychotics, serotonergic drugs).

**E3****ORGANOPHOSPHATE POISONING**

## Purpose

To provide EMTs and Paramedics with a standardised protocol for managing organophosphate poisoning in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing organophosphate poisoning in the prehospital setting.

#### 1.2 Definition

Organophosphate (OP) poisoning is a toxic syndrome resulting from exposure to organophosphate compounds, which inhibit acetylcholinesterase, leading to excess acetylcholine and overstimulation of muscarinic, nicotinic, and central nervous system receptors. Organophosphates are commonly found in **pesticides**, **insecticides**, and **nerve agents** (e.g. sarin).

**Clinical Features:** Symptoms may include:

- **Muscarinic symptoms:** SLUDGE (Salivation, Lacrimation, Urination, Defecation, Gastrointestinal distress, Emesis)
- **Nicotinic symptoms:** Muscle weakness, fasciculations, paralysis
- **CNS symptoms:** Anxiety, confusion, seizures, coma

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Remove the patient from the hot environment.
- **Primary Survey (ABCDE)**
  - **Airway:** Assess and maintain airway patency.
  - **Breathing:** Monitor for signs of respiratory distress, including tachypnoea, laboured breathing, or cyanosis.
  - **Circulation:** Check for hypotension, bradycardia, and signs of shock.
  - **Disability:** Evaluate mental status, looking for confusion, seizures, or coma.
  - **Exposure:** Remove the patient from the source of exposure (if possible). Ensure safety when handling contaminated clothing or materials.
- **Exposure history:** Direct or suspected exposure to pesticides, herbicides, or industrial chemicals. Assess if possible suicide attempt.
- **Signs and symptoms:** Onset of symptoms is typically rapid, with the severity depending on the dose and type of exposure.

### 2.2 Focused Assessment

- **Physical Exam:** Focus on detecting signs of poisoning (e.g., pinpoint pupils, increased secretions, muscle twitching, and weakness).

### 2.3 Vital Signs and Monitoring

- Continuously monitor SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

- **Decontamination**
  - **Ensure safety:** Ensure that the provider and the environment are not contaminated.
  - **Personal Protective Equipment (PPE):** Wear gloves, goggles, and other protective clothing to avoid contamination.
  - **Remove contaminated clothing:** Cut away clothing to avoid spreading contaminants.
  - **Rinse the patient:** If possible, rinse the exposed skin and eyes with copious amounts of water for at least 15 minutes to remove pesticide residues.
- **Airway Management**
  - **Positioning:** Keep the patient in a position that maintains airway patency.
  - **Suction:** Be prepared to suction excessive secretions, including vomitus, saliva, or sputum.
  - **Oxygen:** Administer high-flow oxygen via non-rebreather mask or bag-valve mask as necessary, especially if signs of respiratory distress are present.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Advanced airway if unconscious and required
- IV Fluid resuscitation if hypotensive
- **Pharmacologic Management**
  - **Atropine** (First-line treatment):
    - **Indication:** To treat muscarinic symptoms (e.g., excessive salivation, bradycardia).
    - **Dose:**
      - **Initial dose:** 1-2 mg IV/IO (or IM if IV access is not available).
      - **Repeat:** Every 3-5 minutes, up to 6 mg, until symptomatic improvement (e.g., reduction in secretions, normalisation of heart rate).
- Anticonvulsant if seizure persists, treat as per seizure protocol.

#### 4. Key Points for EMTs and Paramedics

- **Paediatric Patients:** Adjust dosages based on weight.
- **Pregnancy:** The benefits of atropine and pralidoxime generally outweigh the risks in pregnant patients.
- Rapid transport to closest emergency department

**E4****SMOKE / CARBON MONOXIDE POISONING**

## Purpose

To provide EMTs and Paramedics with a standardised protocol for managing smoke inhalation and carbon monoxide poisoning in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing smoke inhalation and carbon monoxide poisoning in the prehospital setting.

#### 1.2 Definition

**Smoke Inhalation:** Inhalation of heated gases, toxic chemicals, or particulate matter during a fire or combustion process can lead to thermal and chemical injury of the airway, lungs, and systemic toxicity.

**Carbon Monoxide (CO) Poisoning:** A form of hypoxic injury caused by the inhalation of carbon monoxide gas, which binds to haemoglobin with an affinity 200–250 times greater than oxygen, reducing oxygen delivery to tissues. CO poisoning is a **common and deadly component** of smoke inhalation and may also occur in **non-fire-related environments** (e.g. faulty heating appliances, exhaust fumes).

#### Clinical Features:

- **Smoke inhalation:** Symptoms may include coughing, wheezing, hoarseness, soot in the mouth/nose, singed facial hair, or burns to the face or upper airway.
- **Carbon monoxide poisoning:** Symptoms range from mild (headache, dizziness) to severe (confusion, unconsciousness, seizures, death), depending on the level of CO exposure.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Do not enter the scene if CO poisoning is suspected and there is no fresh air. When possible, remove the patient from the hazardous environment.
- **Primary Survey (ABCDE)**
  - **Airway:** Assess and maintain airway patency. Look for signs of airway obstruction or burns.
  - **Breathing:** Check for signs of respiratory distress such as laboured breathing, cyanosis, or tachypnoea. Consider the need for supplemental oxygen or advanced airway management.
  - **Circulation:** Assess vital signs, including heart rate, blood pressure, and capillary refill. Check for signs of shock or inadequate perfusion.
  - **Disability:** Assess for neurologic symptoms, including confusion, dizziness, or loss of consciousness.
  - **Exposure:** Remove the patient from the toxic environment (if safe to do so). Assess for burns, soot, or singed facial hair, which may indicate inhalation of hot gases or smoke.

### 2.2 Focused Assessment

- **Physical Exam:** Look for signs of carbon monoxide poisoning (cherry-red skin, confusion, altered mental status) and smoke inhalation (respiratory distress, coughing, wheezing).

### 2.3 Vital Signs and Monitoring

- Continuously monitor SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.



### 3. Management

#### 3.1 EMT-Level Management

##### Oxygen Therapy

- **High-Flow Oxygen:** Administer 100% oxygen via a non-rebreather mask immediately to any patient suspected of having CO poisoning, even if symptoms appear mild, irrespective of pulse oximetry reading.

##### Airway Management

- **Positioning:** Position the patient to maintain airway patency and assist with ventilation if necessary.
- **Suction:** Prepare to suction airway secretions if the patient has respiratory distress, vomiting, or obstruction.

##### Smoke Inhalation Management

- **Positioning:** If the patient is in respiratory distress, position them in a manner that facilitates easier breathing.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- **Advanced Airway:** If the patient is unable to maintain their airway or is in respiratory failure, prepare for advanced airway management (e.g., endotracheal intubation).
- **Bronchodilators:** Consider nebulised Salbutamol 5mg for wheezing or bronchospasm.
- **Steroids:** Consider methylprednisolone (125 mg IV) or Hydrocortisone 100mg for significant smoke inhalation with airway oedema or inflammation.
- Analgesia if burns are present

#### 4. Key Points

- **Paediatric Patients:** Children are more vulnerable to CO poisoning due to their smaller size and higher metabolic rate. Adjust doses and therapies appropriately.
- **Pregnant Patients:** CO poisoning can affect the foetus, and pregnant women may require more aggressive treatment, including hyperbaric oxygen therapy.
- **Burns:** Follow the standard burn protocol for patients with burns and smoke inhalation.

**E5****ENVENOMATION**

## Purpose

To provide EMTs and Paramedics with a standardised protocol for managing envenomation in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing envenomation in the prehospital setting.



## 1.2 Definition

Envenomation is the process by which venom is injected into a person through the bite, sting, or other delivery mechanism of a venomous animal, resulting in local or systemic toxic effects.

Venom differs from poison in that it is **actively delivered** (e.g. fangs, stingers), whereas poison is **passively absorbed**.

Below are some common, medically significant causes of envenomation in East Africa;

### Snakes:

- **Black mamba** (*Dendroaspis polylepis*)
  - Neurotoxic venom – rapid onset of symptoms, can cause paralysis and respiratory arrest
- **Puff adder** (*Bitis arietans*)
  - Cytotoxic venom – causes severe local swelling, pain, and tissue necrosis
- **Boomslang** (*Dispholidus typus*)
  - Haemotoxic venom – causes delayed bleeding and coagulopathy
- **Spitting cobras** (*Naja spp.*)
  - Cytotoxic and neurotoxic – can spit venom into eyes, causing pain and blindness
  - Bites may result in tissue necrosis or neurological symptoms

### Scorpions

- Several **Buthidae** species are present, including Parabuthus spp.
- Venom can cause:
  - Severe pain at the sting site
  - Neurotoxic effects: agitation, sweating, hypertension, rarely seizures or respiratory failure in children

## Spiders

- Although few spider bites in East Africa are medically significant, suspected cases may include:
  - **Button spiders** (*Latrodectus spp.*) – related to the black widow; neurotoxic venom can cause muscle cramps and autonomic symptoms
  - **Sac spiders** – bites may cause local necrosis and ulceration

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Ensure responder and patient safety – **DO NOT TRY TO CAPTURE OR IDENTIFY THE SNAKE**
- **#1 Priority: Reassure the patient and get them to lie down, completely immobilised, do not make the patient walk at all. Snake venom is circulated through the lymphatic system, and the skeletal muscle pump system will increase the speed at which the venom circulates.**
- **Primary Survey (ABCDE)**
  - **Airway:** Assess and maintain airway patency. Be aware of swelling or potential airway compromise from severe envenomation, particularly in the face or neck area.
  - **Breathing:** Check for signs of respiratory distress (e.g., stridor, laboured breathing). Provide oxygen as necessary.
  - **Circulation:** Assess for signs of shock, hypotension, or tachycardia. Look for evidence of bleeding or signs of systemic effects such as dizziness, weakness, or altered mental status.
  - **Disability:** Evaluate for neurological symptoms, such as confusion, seizures, or paralysis, which may occur with certain types of venom.
  - **Exposure:** Inspect the bite or sting site for signs of envenomation (puncture marks, swelling, erythema, fang marks, etc.).

## 2.2 Focused Assessment

- **History**
  - **Exposure history:** If possible, obtain a description of the animal (size, colour, appearance).
  - **Time of exposure:** Determine when the bite occurred and what the patient was doing at the time.
- **Symptoms:** Ask about any immediate pain, swelling, dizziness, weakness, or systemic symptoms (nausea, vomiting, difficulty breathing).
- **Physical Exam:** Focus on the area of envenomation, assessing for swelling, redness, bruising, or necrosis. In the case of snake bites, look for fang marks and signs of local or systemic envenomation.

## 2.3 Vital Signs and Monitoring

- Continuously monitor SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.

### 3. Management

#### 3.1 EMT-Level Management

- **Stay calm:** Ensure the patient is calm and reassured. Anxiety can exacerbate the effects of envenomation.
- **Remove the patient from danger:** Ensure that the patient is no longer at risk of further bites.
- **Positioning:** Keep the patient calm and in a position that minimises movement of the affected limb (preferably at or below the level of the heart to slow venom spread).
- **Pressure Immobilisation Technique:** Bandage the affected limb from as distal as possible up to the most proximal end with a firm conforming bandage; it should not arrest circulation but provide firm pressure to the limb. Also, splint the limb using a wooden pole or splint.
- **Do not:** suck the venom, apply ice, promote bleeding or cut the bite.
- **Remove tight clothing around the site due to swelling**
- **Monitor and reassess:** Monitor vital signs continuously and reassess the patient's condition regularly. Be prepared for sudden changes, particularly in cases of severe envenomation.

#### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Advanced airway if the patient is unconscious and required
- Anticonvulsant if seizure persists, treat as per seizure protocol.
- IV Fluid resuscitation if hypotensive

#### 4. Key Points

- **Transport Priority: Rapid transport** is essential, once the patient is immobilised and PIT bandage applied, rapid, emergency transport with lights and sirens should be undertaken **directly to a facility that has antivenom** if the transport time is less than 1 hour and the patient is not peri-arrest.
- Early administration of **polyvalent antivenom**, where available, is crucial
- **Monitor en route:** Continuously monitor vital signs, including heart rate, blood pressure, respiratory rate, and oxygen saturation. Be prepared for anaphylaxis, respiratory distress, or other complications.
- **Notify receiving facility:** Provide information about the type of envenomation and any treatments provided so the hospital can prepare accordingly.
- **Paediatric Patients:** Children may have more severe reactions to envenomation due to their smaller size. Remember that a child, whilst smaller, has received the same amount of venom that an adult would and, therefore, should receive the same dose of antivenom that an adult would.
- **Pregnancy:** Antivenom can have serious side effects in pregnancy. Studies have found that the foetal death rate is between 56% and 58% when mothers receive antivenom. The most senior healthcare provider must make a clinical decision as to whether the patient should be monitored or administered antivenom.



**E6****TRADITIONAL MEDICINE / HERBAL EMERGENCIES**

## Purpose

To provide EMTs and Paramedics with a standardised protocol for managing traditional medicine emergencies in prehospital settings.

### 1. Scope and Definitions

#### 1.1 Scope

Applicable to EMTs and Paramedics managing traditional medicine emergencies in the prehospital setting.

#### 1.2 Definition

A traditional medicine emergency is a medical crisis or acute deterioration in a patient's condition that results from, or is significantly influenced by, the use of traditional or non-biomedical therapies—either through direct adverse effects, delay in accessing appropriate care, or harmful interactions with conventional treatments.

Traditional medicine practices vary significantly across cultures and may include the use of herbal remedies, animal-based products, acupuncture, and spiritual or ritualistic practices. While many of these treatments are considered safe when used appropriately, adverse effects and medical emergencies can occur, particularly when traditional therapies are used incorrectly or in place of clinical care during serious or life-threatening conditions. It is common for patients to first seek help from traditional healers and only present to formal healthcare services when their condition worsens or fails to improve with traditional treatments.

Emergencies related to traditional medicine may include:

- **Herbal Toxicity:** Poisoning from plants or plant-based compounds.
- **Acupuncture Complications:** Infections, bleeding, or pneumothorax from improperly administered needles.
- **Unregulated Supplements:** Overdose or allergic reactions from unregulated or contaminated supplements.
- **Spiritual or Psychological Emergencies:** Psychological distress or physical symptoms triggered by traditional spiritual practices or beliefs.

## 2. Assessment

### 2.1 Initial Assessment

- **Scene Safety:** Consider responder and patient safety.
- **Primary Survey (ABCDE)**
  - **Airway:** Ensure the patient's airway is clear and assess for signs of obstruction, especially in cases of severe allergic reactions or swelling from herbal ingestion or acupuncture.
  - **Breathing:** Monitor respiratory status, especially if there are signs of anaphylaxis, airway compromise, or pneumothorax (from acupuncture complications).
  - **Circulation:** Monitor vital signs closely, paying attention to hypotension, tachycardia, or signs of shock that may suggest systemic toxicity.
  - **Disability:** Assess mental status for signs of confusion, altered consciousness, or psychological distress, especially if linked to spiritual or mental health practices.
  - **Exposure:** Inspect for signs of poisoning (e.g., skin rashes, burns, swelling, or injection site issues) or any potential complications related to traditional treatments (e.g., acupuncture needle site infection).

## 2.2 Focused Assessment

- **History**
  - **Traditional Medicine Use:** Ask if the patient has used any traditional or alternative remedies (herbal medicines, acupuncture, spiritual practices, etc.) in the recent past.
    - **Herbal Remedies:** Be specific about the names of the herbs, plants, or substances used.
    - **Acupuncture:** Inquire if the patient has recently undergone acupuncture treatment and whether they have any concerns or symptoms related to the procedure.
    - **Spiritual Practices:** Ask about involvement in any spiritual practices (e.g., rituals, energy healing) that may have contributed to their symptoms.
- **Physical Exam:** Focus on examining any puncture sites, signs of infection, or physical evidence of poisoning (e.g., rash, nausea, vomiting, diaphoresis, etc.).
- **Psychological Assessment:** If relevant, assess the patient for signs of psychological distress, such as paranoia, confusion, or hallucinations, which may result from spiritual or psychological practices.

## 2.3 Vital Signs and Monitoring

- Continuously monitor SPO<sub>2</sub>, respiratory rate, blood pressure, heart rate, temperature, and blood glucose.
- Assess for signs of anaphylaxis or systemic toxicity.

### 3. Management

#### 3.1 EMT-Level Management

- **General Management Principles**
  - **Reassurance and Education:** Calm and reassure the patient (and family) while explaining that their safety and well-being are the primary focus. Ensure that the patient feels comfortable sharing their use of traditional medicine to avoid delayed treatment.
  - **Safety First:** Take precautions when handling substances or materials associated with traditional medicine (e.g., herbal products, acupuncture needles).
  - **Monitoring and Observation:** Continuously monitor the patient's vital signs, including heart rate, respiratory status, and oxygen saturation. Be prepared to treat any complications that arise.
- **Herbal Toxicity**
  - **Initial Management:** If herbal toxicity is suspected (e.g., from ingestion of a poisonous plant or excessive use of a medicinal herb), provide supportive care:
    - **Fluid Resuscitation:** Provide IV fluids (e.g., normal saline or lactated Ringer's) to maintain circulation and hydration.
    - **Activated Charcoal:** If the herb was ingested within the last hour, consider administering activated charcoal to limit further absorption.
  - **Common Toxic Herbs:** Examples include Aconite, Foxglove (Digitalis), Jimsonweed (Datura), and Pennyroyal.
- **Acupuncture Complications**
  - **Infection:** If signs of infection (redness, swelling, pus) are observed at the acupuncture needle insertion sites:
    - Clean the area with sterile saline.
    - Antibiotics: Consider administering systemic antibiotics if there are signs of systemic infection or cellulitis.
  - **Pneumothorax:** If the patient develops respiratory distress or signs of pneumothorax (e.g., sudden chest pain, shortness of breath), prepare for immediate transport and possible chest decompression if pneumothorax tensions.
  - **Needle Retention:** If a needle is retained at the acupuncture site, stabilise the needle and transport the patient to the hospital for removal.

- **Unregulated Supplements or Overdose**
  - **Initial Management:** If the patient has taken unregulated supplements or overdosed on over-the-counter products (e.g., herbal supplements, vitamins, etc.), initiate treatment based on symptoms:
    - **Oxygen:** Administer high-flow oxygen as needed, especially if there is any respiratory distress.
    - **IV Fluids:** Provide IV fluids to support circulation, especially if the patient is in shock.
    - **Activated Charcoal:** If indicated, administer activated charcoal if the substance was ingested recently.
- **Psychological or Spiritual Emergencies**
  - **Calm the Patient:** If the patient is experiencing distress from psychological or spiritual practices, provide a calm and reassuring environment. Avoid judgmental or confrontational behaviour.

### 3.2 Paramedic-Level Management

- **EMT-Level Management +**
- Advanced airway if unconscious and required
- Anticonvulsant if seizing persists, treat as per seizure protocol.
- Sedation if required, follow Acute Behavioural Disturbance protocol.
- **Herbal Toxicity**
  - **Fluid Resuscitation:** Provide IV fluids (e.g., normal saline or lactated Ringer's) to maintain circulation and hydration.
  - **Activated Charcoal:** If the herb was ingested within the last hour, consider administering activated charcoal to limit further absorption.
- **Acupuncture Complications**
  - **Pneumothorax:** If the patient develops respiratory distress or signs of pneumothorax (e.g., sudden chest pain, shortness of breath), prepare for immediate transport and possible chest decompression if pneumothorax tensions.
- **Unregulated Supplements or Overdose**
  - **IV Fluids:** Provide IV fluids to support circulation, especially if the patient is in shock.
  - **Activated Charcoal:** If indicated, administer activated charcoal if the substance was ingested recently.

#### 4. Key Points

- Rapid transport to the closest emergency department if altered physiology or ABC derangement.
- For patients experiencing psychological or psychiatric crises or exhibiting dangerous behaviour, prepare for transport to a medical facility capable of providing psychiatric care if possible.
- If the patient lacks mental capacity, treatment must be given in the patient's best interest.

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